AD 2. AERODROMES

VAAH AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VAAH - SARDAR VALLABHBHAI PATEL INTERNATIONAL AIRPORT, AHMEDABAD

VAAH AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	230416N 0723735E M. Brg. 035.20 DEG/ 712M from physical extremity of RWY 05	
2	Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves	045 DEG/8KM from Ahmedabad Railway station	
3	Aerodrome elevation and reference temperature	189 FT / 42.4 DEG C	
4	Magnetic variation, date of information and annual change	0.25 DEG W (2010) /0.033 DEG E	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	Ahmedabad International Airport Ltd, Chief Airport Officer 1st Floor, Terminal 1, Sardar Vallabhbhai Patel International Airport, Hansol, Ahmedabad, Gujarat - 380003,	
		Telephone:	917925554790 918980015105 916358941440
		Fax:	
		AFS:	---
		Email:	cao.ahmedabadairport@adani.com
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	

7	Remarks	1.Website: https://ahmedabad.adaniairports.com/ 2. Elevation in EGM08 3. Non scheduled aircraft operators are required to file their slot request on aocc.amd@adani.com for clearance. All non scheduled/ General Aviation movements will be subjected to positive approval from Airport operations control centre (AOCC). Request for International Movement must be forwarded at least 48hrs in advance. Request for Domestic movement must be forwarded at least 24 hrs in advance. Aerodrome reference code 4E AOCC Mobile No +91 6358877435 4. All Scheduled aircraft operators are required to file their slot request on slot.amd@adani.com

VAAH AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	MON-FRI 0400-1300 UTC (0930-1830 IST), Sat 0400-0800 UTC (0930-1330 IST) Sun/Holiday: Nil
2	Custom and immigration	H24
3	Health and sanitation	H24
4	AIS briefing office	H24
5	ATS reporting office (ARO)	H24
6	MET Briefing office	H24
7	Air Traffic Service	H24
8	Fuelling	H24
9	Handling	AIASL (Air India Airport Services limited), GSEC Bird Airport services Pvt Ltd
10	Security	H24
11	De-icing	NIL

12	Remarks	The approved hourly RWY traffic handling capacity: Maximum number of arrival and departures -20 Maximum number of arrivals only -15 Maximum number of departures only -12
		All helicopters and aircraft operator from uncontrolled airfields within VAAH TMA shall file FPL or online FPL portal https://www.onlinefpl.in and obtain ADC, FIC No. and ATC CLR FM VABB FIC. Prior ATC CLR BFR DEP FM VAAH if departing from airfield within 15 NM of VAAH on TEL NR 079-22869248. After DEP CTC VAAH APP on VHF 119.8MHZ.

VAAH AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	Upto B747-400 Manual by arrangement with local Airlines. 1) Facilities: Export Perishable Terminal, Import Cold Zone, Export Unitization Zone, Export Heavy and Bonded Cargo Terminal, Cold Storage Facilities, Dangerous Goods Storage, Valuable & Vulnerable Cargo Storage, General export storage. 2) Handling Equipment: Forklifts (10T/5T/3T), Weighing Scales (30T / 5T / 3T/ 1T), Hydraulic Pallet Trucks, Truck Docks, Dock Levelers, Platform Trolleys, Workstations for ULD Buildup, Multilevel Racking Systems, dollies, and slave pallets. 3) IT/Systems: Dedicated Cargo Management System, Air Cargo Community Portal (GMAX), Cargo Mobile Application. 4) Parking system for heavy vehicles: Dedicated facilities for parking of heavy vehicles, and truck management system avbl.
2	Fuel and Oil types	JET A1
3	Fuelling facilities and capacity	M/s. ADANI AVIATION FUEL SERVICES LIMITED No. of Storage tanks: 10 Capacity: 2800 KL No. of refuelers: 17 refuelers S. No. Capacity 1 16 KL 15 2 25 KL 2 Total 17 Duty Mobile No . +91 7777960933
4	De-icing facilities	NIL
5	Hangar space for visiting aircraft	Not Available
6	Repair facilities for visiting aircraft	LIMITED, PRIOR ARRANGEMENT WITH LOCAL AIRLINES. IN ADDITION, MRO (AIESL, AIRWORKS AND INDAMER HAVE REPAIR FACILITIES FOR GENERAL AVIATION AIRCRAFTS. AIESL : DUTY CELL NO.+91 9825600231
7	Remarks	NON SKD aircraft should ensure availability of tow bar on board or with their ground handling agent

VAAH AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	In the city & near the AD.

2	Restaurant(s) at or in the vicinity of aerodrome	At AD and in the city
3	Transportation possibilities	Taxi, Car hire from AD, train, BRTS buses from the city, Self drive car, Auto, Intercity buses
4	Medical Facilities	First aid at AD. Doctor on call, hospitals in the city
5	Bank and post office at or in the vicinity of aerodrome	Banks: ATMs at AD H24 Post office: Limited hours at Domestic Terminal
6	Tourist office	During flight timings at domestic terminal
7	Remarks	Currency Exchange Counter available at Intl and Domestic Terminal.

VAAH AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-9
2	Rescue equipment	AVBL as per category.
3	Capability for removal of disabled aircraft	1. The Critical Aircraft of Airport is A350. 2. Primary Responsibility for removal of disabled aircraft rests with the concerned airline. 3. The airlines may contact 'Aerodrome Coordinator for Disabled Aircraft Removal Operations 'for coordinating with the agencies having Disabled Aircraft Recovery Kit. 4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations: Aerodrome Coordinator: Head Aero Operations AIAL Email: aocc.amd@adani.com,iaocahmedabad@adani.com Contact No.: +91 9426206472
4	Remarks	Nil

VAAH AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	Manual and Mechanical Sweeper
2	Clearance priorities	NIL
3	Remarks	NIL

VAAH AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Designation, surface and strength of aprons	Refer Aircraft Parking / Docking Charts for Details
2	Designation, width, surface and strength of taxiways	Refer Aircraft Parking / Docking Chart for Details and AD2.23
3	Location and elevation of altimeter checkpoints	Location: All aircraft Stands Elevation: Refer Aircraft Parking/Docking Chart
4	Location of VOR checkpoints	TWY B: 230404.03N0723722.72E/182FT TWY H: 230429.56N0723742.21E/181 FT Elevations in EGM08
5	Position of INS checkpoints	NIL

<table>
<tr><td>6</td><td>Remarks</td><td>
1. Apron Details
<table>
<tr><td>Designation</td><td>Apron-1 (Domestic)</td><td>Apron-2 (International)</td><td>Apron-3</td></tr>
<tr><td>Location</td><td>North-West of RWY 05 beginning</td><td>Northwest of RWY 05 connected via TWY G and H----</td><td>North-west of Apron 1 connected via Taxiway L5</td></tr>
<tr><td>Dimension</td><td>780M X200M</td><td>380M X 136M</td><td>150M X 120M</td></tr>
<tr><td>Shoulder</td><td>10.5M</td><td>10.5M</td><td>3.5M all sides</td></tr>
<tr><td>Light</td><td>Blue edge lights, flood lights.</td><td>Blue edge lights, flood lights.</td><td>Edge lights, flood lights.</td></tr>
<tr><td>2.Isolated Aircraft Parking Position:</td><td colspan="3">110 X 91 M
Surface: Concrete,
PCR: 354/R/D/W/T,
Blue edge Lights.
COORD / ELEV:
230352.15278N 0723732.49080E / 184 FT</td></tr>
</table>
3. Coordinates of Runway Holding Position:

i. TWY A: 230359.78528N 0723718.64727E/180FT
ii. TWY B: 230403.79991N 0723722.96814E/182FT
iii. TWY C: 230417.80491N 0723737.15210E/181FT
iv. TWY D: 230437.02077N 0723757.41346E/183FT
v. TWY F: 230359.58749N 0723727.74313E/184FT
vi. TWY K: 230355.41597N 0723722.96961E/181FT
vii. TWY R4: 230442.580 N 0723804.420 E/185FT
viii) TWY R1: 23 0508.76N 723832.42E/ 187 FT
ix) TWY R1A: 230520.331N 0723844.779E/188 FT
4. Coordinates of RWY & Taxi Intersection:

i. TWY A: 230357.71594N 0723720.95119E/180FT
ii. TWY B: 230401.70483N 0723725.21235E/182FT
iii. TWY C: 230419.66822N 0723744.40310E/183FT
iv. TWY D: 230438.89346N 0723804.94412E/185FT
v. TWY F: 230401.74697N 0723725.25737E/182FT
vi. TWY K: 230357.93834N 0723721.18877E/180FT
vii. TWY R4: 23 04 40.593N 072 38 06.694E
viii) TWY R1: 230506.70N 0723834.68E/ 189FT
ix) TWY R1A: 230518.274N 0723847.028E/189 FT

5. Access to Isolated Aircraft Parking Position is through TWY F. Aircraft shall be taken to Isolated Aircraft Parking Position under ‘Follow me’

6. Gravity Model: EGM 08
</td></tr>
</table>

VAAH AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	The stands have been provided with illuminated stand identification boards with respective coordinates displayed.Illuminated stand identifications boards on stand 70L,71L,71, 71R 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 31A, 31L, 31, 31R, 32L, 32, 32R, 33, 34L,34, 35, 36L, 36, 36R, 37, 38, 39, 40, 41, 42 and 43. Retroreflective Stand identification board provided on stand 34R. Stand identification boards not available for stands 70, 70R,25, 26, 27, 28, 29, 30, 25A, 26A, 27A, 28A, 29A, and 30A. However, stand identification markings are provided on ground. Taxing guidance signs are provided on all taxiway intersections and straight sections. VDGS/AVDGS provided on stands 77, 78, 79, 80, 31A, 31L, 31, 31R, 32L, 32, 32R, 33, 34 34L and 35 with aerobridge facility and available for perations H24.
2	Runway and taxiway markings and lights	RWY Markings: Designation, THR, TDZ, Centreline, Aiming point, Side stripe, Turn Pad Lights: THR,RWY Edge, RWY End, Turn pad, Wing bar lights(Runway 23) TWY Marking: Centreline, Runway, Holding Positions, TWY, Intermediate holding position, Transverse Stripe,Side stripe, Maxspan, No-entry markings (provided on Taxiway R3 and R5). Lights: Taxiway Edge lights Avbl on A, B, C, D, F, G, H, K, M, P, R4, R, R1, R3, R5, R1A. No entry bar, Taxiway Centre Line Lights provided for RET R5, “MAXSPAN 36M” provided on Taxiway R, R1, R3, R1A. Runway Guard Lights configuration “A” available at A, B, C, D, R4,R1, R3, R5, R1A.
3	Stop bars (if any)	NIL
4	Remarks	All aircraft taxing via apron 1 use minimum power in order to avoid jet blast..

VAAH AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
23/APCH 05/TKOF	FENCE	230524.4N 0723850.7E	205 FT	LGTD	Boundary Wall with Fencing on Top
23/APCH 05/TKOF	FENCE	230525.2N 0723849.9E	201 FT	LGTD	Boundary Wall with Fencing on Top
23/APCH 05/TKOF	FENCE	230526.4N 0723848.5E	201 FT	LGTD	Boundary Wall with Fencing on Top
23/TKOF 05/APCH	FENCE	230526.1N 0723848.9E	200 FT	LGTD	Boundary Wall with Fencing on Top
23/APCH 05/TKOF	FENCE	230525.6N 0723849.4E	204 FT	LGTD	Boundary Wall with Fencing on Top
23/APCH 05/TKOF	FENCE	230525.5N 0723849.5E	204 FT	LGTD	Boundary Wall with Fencing on Top
23/APCH 05/TKOF	FENCE	230523.9N 0723851.2E	207 FT	LGTD	Boundary Wall with Fencing on Top

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
23/TKOF 05/APCH	BUILDING	230351.5N 0723709.2E	201 FT	LGTD	GAGAN INRES Room, Antenna Top: 61.13 M Antenna Top: 60.99 M, Antenna Top: 60.11 M, Sintex Tank Top: 59.89 M, Grill Top: 59.40 M, Building Top: 58.43 M.
23/APCH 05/TKOF	FENCE	230522.4N 0723852.7E	203 FT	LGTD	Boundary Wall with Fencing on Top
23/APCH 05/TKOF	FENCE	230520.8N 0723854.3E	203 FT	LGTD	Boundary Wall with Fencing on Top
23/APCH 05/TKOF	FENCE	230520.0N 0723855.1E	203 FT	LGTD	Boundary Wall with Fencing on Top
23/APCH 05/TKOF	FENCE	230519.5N 0723855.5E	203 FT	LGTD	Boundary Wall with Fencing on Top
23/APCH 05/TKOF	FENCE	230519.2N 0723855.8E	205 FT	LGTD	Boundary Wall with Fencing on Top
23/APCH 05/TKOF	OTHER	230522.3N 0723852.6E	199 FT	LGTD	Security Hut
23/APCH 05/TKOF	OTHER	230520.5N 0723848.4E	205 FT	NO	Mobile Road Traffic (Road Elev - 57.55m + 5m Mobile Traffic)
23/APCH 05/TKOF	OTHER	230521.8N 0723849.6E	206 FT	NIL	Mobile Road Traffic (Road Elev - 57.85m + 5m Mobile Traffic)
23/APCH 05/TKOF	OTHER	230523.5N 0723851.2E	208 FT	NIL	Mobile Road Traffic (Road Elev - 58.26m + 5m Mobile Traffic)
23/APCH 05/TKOF	OTHER	230523.0N 0723851.7E	208 FT	NIL	Mobile Road Traffic (Road Elev - 58.33m + 5m Mobile Traffic
23/APCH 05/TKOF	OTHER	230522.4N 0723852.4E	208 FT	NIL	Mobile Road Traffic (Road Elev - 58.40m + 5m Mobile Traffic)
23/APCH 05/TKOF	OTHER	230521.7N 0723853.1E	208 FT	NIL	Mobile Road Traffic (Road Elev - 58.38m + 5m Mobile Traffic)
23/APCH 05/TKOF	OTHER	230524.1N 0723850.6E	207 FT	NIL	Mobile Road Traffic (Road Elev - 58.08m + 5m Mobile Traffic)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
23/TKOF 05/APCH	BUILDING	230343.2N 0723711.3E	214 FT	NIL	Building, Parapet Wall 65.15 M, Slab Top: 64.50 M.
23/TKOF 05/APCH	BUILDING	230343.1N 0723711.2E	214 FT	NIL	Building, Building Top: 65.19 M
23/TKOF 05/APCH	BUILDING	230342.6N 0723711.0E	213 FT	NIL	Building Top
23/TKOF 05/APCH	BUILDING	230342.5N 0723711.0E	213 FT	NIL	Building, Staircase Room Top: 65.06 M, Building Top: 63.88 M.
23/TKOF 05/APCH	BUILDING	230341.3N 0723713.3E	211 FT	NIL	Building, Staircase Room Top: 64.22 M, Building Top: 62.56 M.
23/TKOF 05/APCH	BUILDING	230341.9N 0723713.0E	212 FT	NIL	Building Top
23/TKOF 05/APCH	BUILDING	230341.6N 0723713.8E	212 FT	NIL	Building Top
23/APCH 05/TKOF	TREE	230519.4N 0723855.7E	219 FT	NIL	Tree
23/TKOF 05/APCH	TREE	230344.6N 0723655.7E	250 FT	NIL	Group of Trees
23/TKOF 05/APCH	TREE	230338.9N 0723652.5E	253 FT	NIL	Group of Trees
23/TKOF 05/APCH	TREE	230339.3N 0723649.8E	245 FT	NIL	GROUP OF TREES
23/TKOF 05/APCH	TREE	230339.0N 0723654.9E	248 FT	NIL	Group of Trees
23/TKOF 05/APCH	TREE	230338.8N 0723655.0E	247 FT	NIL	TREE
23/TKOF 05/APCH	OTHER	230356.9N 0723712.5E	193 FT	NIL	Mobile Road Traffic (Road Elev: 53.97 M + 5 M Mobile Traffic)
23/APCH 05/TKOF	OTHER	230522.8N 0723851.9E	208 FT	NIL	Mobile Road Traffic (Road Elev: 58.29 M + 5 M Mobile Traffic)
23/APCH 05/TKOF	OTHER	230519.1N 0723855.6E	208 FT	NIL	Mobile Road Traffic (Road Elev: 58.49 M + 5 M Mobile Traffic)
23/APCH 05/TKOF	FENCE	230526.8N 0723848.2E	199 FT	NIL	Airport Boundary Wall with Fencing on Top
23/APCH 05/TKOF	BUILDING	230520.2N 0723856.0E	210 FT	NIL	Temple
23/TKOF 05/APCH	BUILDING	230343.2N 0723711.9E	213 FT	NIL	House
23/APCH 05/TKOF	BUILDING	230522.0N 0723854.2E	204 FT	NIL	House

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
23/APCH 05/TKOF	BUILDING	230521.4N 0723854.6E	203 FT	NIL	House
23/APCH 05/TKOF	BUILDING	230521.0N 0723854.8E	204 FT	NIL	House
23/APCH 05/TKOF	BUILDING	230520.5N 0723855.1E	204 FT	NIL	House
23/APCH 05/TKOF	BUILDING	230519.7N 0723855.6E	202 FT	NIL	House
23/APCH 05/TKOF	BUILDING	230522.3N 0723854.0E	204 FT	NIL	House
23/APCH 05/TKOF	BUILDING	230523.3N 0723854.5E	203 FT	NIL	HOUSE
23/APCH 05/TKOF	BUILDING	230523.5N 0723854.7E	203 FT	NIL	House
23/TKOF 05/APCH	TREE	230347.9N 0723659.9E	227 FT	NIL	GROUP OF TREES
23/TKOF 05/APCH	TREE	230346.2N 0723658.9E	239 FT	NIL	GROUP OF TREES
23/APCH 05/TKOF	BUILDING	230637.0N 0723949.1E	353 FT	LGTD	Kailash Rejoice 2, Antenna Top: 107.46 M, Pole Top: 107.18 M, Staircase Room Top: 105.45 M, Building Top: 104.12 M.
23/APCH 05/TKOF	BUILDING	230636.8N 0723947.5E	346 FT	LGTD	Kailash Rejoice 2, Staircase Room Top: 105.46 M, Building Top: 103.76 M.
In circling area and at AD	OTHER	230412.4N 0723731.5E	190 FT	NIL	Bird Scarer Hut
In circling area and at AD	OTHER	230409.0N 0723729.8E	186 FT	NIL	PAPI
In circling area and at AD	OTHER	230408.8N 0723730.1E	186 FT	NIL	PAPI
In circling area and at AD	OTHER	230408.6N 0723730.3E	186 FT	NIL	PAPI
In circling area and at AD	OTHER	230408.4N 0723730.5E	186 FT	NIL	PAPI
In circling area and at AD	OTHER	230416.7N 0723738.8E	186 FT	NIL	Sign Board
In circling area and at AD	BUILDING	230425.0N 0723743.9E	185 FT	LGTD	Temple
In circling area and at AD	OTHER	230424.9N 0723745.0E	187 FT	NIL	Bird Scarer Hut
In circling area and at AD	OTHER	230501.1N 0723823.3E	193 FT	NIL	Bird Scarer Hut
In circling area and at AD	OTHER	230451.7N 0723824.1E	194 FT	NIL	Bird Scarer Hut
In circling area and at AD	OTHER	230511.9N 0723834.8E	209 FT	LGTD	Windsock

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	230513.4N 0723835.7E	200 FT	LGTD	Glide Path Hut, Obstruction Light Top: 61.06 M Hut Top: 61.03 M.
In circling area and at AD	POLE	230513.5N 0723834.6E	203 FT	NIL	Antenna Top
In circling area and at AD	BUILDING	230514.2N 0723834.9E	201 FT	LGTD	Hut, Pipe Top: 61.29 M, Hut Top: 60.22 M.
In circling area and at AD	OTHER	230513.2N 0723835.6E	196 FT	NO	Transmitter/ Receiver
In circling area and at AD	OTHER	230512.5N 0723834.8E	196 FT	NIL	Transmitter/ Receiver
In circling area and at AD	OTHER	230511.3N 0723834.3E	196 FT	NIL	Bird Scarer Hut
In circling area and at AD	OTHER	230510.8N 0723832.7E	189 FT	NIL	Electric Box
In circling area and at AD	SIGN	230513.4N 0723837.0E	190 FT	NIL	Sign Board (Stop)
In circling area and at AD	OTHER	230512.5N 0723834.7E	191 FT	NIL	Electric Box
In circling area and at AD	OTHER	230512.6N 0723834.7E	191 FT	NIL	Electric Box
In circling area and at AD	OTHER	230513.2N 0723835.7E	193 FT	NIL	Electric Box
In circling area and at AD	OTHER	230512.4N 0723846.4E	196 FT	NIL	Bird Scarer Hut
In circling area and at AD	POLE	230526.9N 0723847.8E	217 FT	LGTD	Communication Antenna, (Ru09) Antenna Top: 66.25 M, Obstruction Light Top: 65.81 M
In circling area and at AD	OTHER	230405.6N 0723736.0E	205 FT	LGTD	Windsock, Obstruction Light Top: 62.45 M Windsock Top: 60.30 M.
In circling area and at AD	OTHER	230356.1N 0723719.2E	182 FT	NIL	Approach Light
In circling area and at AD	SIGN	230435.8N 0723757.2E	185 FT	NIL	Sign Board
In circling area and at AD	SIGN	230435.9N 0723757.7E	185 FT	NIL	Sign Board
In circling area and at AD	SIGN	230438.7N 0723758.4E	186 FT	NIL	Sign Board
In circling area and at AD	SIGN	230441.6N 0723805.5E	187 FT	NO	Sign Board (D)
In circling area and at AD	OTHER	230447.7N 0723809.0E	190 FT	NIL	Bird Scarer Hut
In circling area and at AD	SIGN	230455.4N 0723817.6E	187 FT	NIL	Sign Board (Stop)
In circling area and at AD	POLE	230456.9N 0723816.3E	199 FT	LGTD	Communication Antenna Top (Ru10)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	SIGN	230357.1N 0723722.6E	184 FT	NIL	Sign Board
In circling area and at AD	SIGN	230358.9N 0723717.7E	183 FT	NIL	Sign Board
In circling area and at AD	SIGN	230359.0N 0723717.7E	183 FT	NIL	Sign Board
In circling area and at AD	SIGN	230359.2N 0723717.4E	183 FT	NIL	Sign Board
In circling area and at AD	SIGN	230401.1N 0723722.1E	183 FT	NIL	Sign Board
In circling area and at AD	SIGN	230401.3N 0723722.4E	184 FT	NIL	Sign Board
In circling area and at AD	SIGN	230401.1N 0723719.3E	183 FT	NIL	Sign Board
In circling area and at AD	SIGN	230402.5N 0723719.2E	182 FT	NIL	Sign Board
In circling area and at AD	SIGN	230402.9N 0723722.0E	184 FT	NIL	Sign Board
In circling area and at AD	SIGN	230403.0N 0723721.9E	184 FT	NIL	Sign Board
In circling area and at AD	SIGN	230404.2N 0723724.6E	185 FT	NIL	Sign Board
In circling area and at AD	SIGN	230404.7N 0723723.8E	185 FT	NIL	Sign Board
In circling area and at AD	SIGN	230405.1N 0723723.7E	185 FT	NIL	Sign Board
In circling area and at AD	SIGN	230407.0N 0723723.8E	185 FT	NIL	Sign Board
In circling area and at AD	FENCE	230518.9N 0723856.1E	205 FT	LGTD	Boundary Wall with Fencing on Top
In circling area and at AD	POLE	230518.6N 0723855.7E	210 FT	LGTD	Communication Antenna (ASMGCS, Ru08)
In circling area and at AD	OTHER	230512.6N 0723849.3E	206 FT	LGTD	Watch Tower
In circling area and at AD	POLE	230508.8N 0723842.8E	206 FT	NIL	Met Antenna, Wind Direction Indicator: 62.81M Wind Speed Indicator: 62.67 M Antenna Top: 62.51 M.
In circling area and at AD	POLE	230508.7N 0723842.9E	197 FT	LGTD	Obstruction Light
In circling area and at AD	OTHER	230439.6N 0723809.2E	197 FT	NIL	Bird Scarer Hut
In circling area and at AD	SIGN	230437.0N 0723807.9E	188 FT	NIL	Sign Board
In circling area and at AD	POLE	230436.8N 0723810.0E	191 FT	NIL	Light Pole
In circling area and at AD	POLE	230438.4N 0723811.1E	190 FT	NIL	LIGHT POLE
In circling area and at AD	POLE	230436.5N 0723808.7E	188 FT	NIL	LIGHT POLE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	POLE	230436.6N 0723808.2E	188 FT	NIL	Light Pole
In circling area and at AD	POLE	230435.6N 0723807.4E	188 FT	NIL	Light Pole
In circling area and at AD	POLE	230434.8N 0723806.9E	188 FT	NIL	Light Pole
In circling area and at AD	POLE	230434.0N 0723806.3E	188 FT	NIL	Light Pole
In circling area and at AD	POLE	230433.2N 0723805.7E	188 FT	NIL	Light Pole
In circling area and at AD	POLE	230432.8N 0723805.5E	188 FT	NIL	Light Pole
In circling area and at AD	POLE	230440.0N 0723812.3E	190 FT	NIL	Light Pole
In circling area and at AD	POLE	230437.3N 0723810.4E	191 FT	NIL	Light Pole
In circling area and at AD	POLE	230439.5N 0723811.9E	190 FT	NIL	Light Pole
In circling area and at AD	POLE	230440.2N 0723813.1E	191 FT	NIL	Light Pole
In circling area and at AD	POLE	230440.4N 0723812.9E	191 FT	NIL	Light Pole
In circling area and at AD	OTHER	230459.6N 0723832.7E	196 FT	NIL	Bird Scarer Hut
In circling area and at AD	OTHER	230507.4N 0723838.6E	190 FT	NIL	PAPI
In circling area and at AD	OTHER	230507.6N 0723838.3E	190 FT	NIL	PAPI
In circling area and at AD	OTHER	230507.8N 0723838.1E	190 FT	NIL	PAPI
In circling area and at AD	OTHER	230508.0N 0723837.9E	190 FT	NIL	PAPI
In circling area and at AD	POLE	230449.6N 0723824.6E	214 FT	LGTD	Communication Antenna (Ru07)
In circling area and at AD	OTHER	230519.9N 0723848.7E	191 FT	NIL	Approach Light
In circling area and at AD	OTHER	230519.2N 0723848.0E	191 FT	NIL	Approach Light
In circling area and at AD	POLE	230508.8N 0723842.9E	206 FT	NIL	MET Antenna, Wind Speed Indicator Located North Side of RWY 23.
In circling area and at AD	POLE	230508.8N 0723842.8E	205 FT	NIL	MET Antenna Top Located North Side of RWY 23.
In circling area and at AD	POLE	230508.7N 0723842.9E	205 FT	LGTD	Met Antenna, Wind Speed Indicator Top Located North Side of RWY 23.

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	230441.3N 0723804.0E	187 FT	NIL	Landing-T
In circling area and at AD	BUILDING	230501.5N 0723839.6E	237 FT	NIL	Building, OHT: 72.38 M, Staircase Room Top: 71.31 M Parapet Wall: 69.80 M, Building Slab Top: 68.82 M.
In circling area and at AD	BUILDING	230502.0N 0723840.1E	237 FT	NIL	Building, OHT: 72.38 M, Staircase Room Top: 71.31 M, Parapet Wall: 69.80 M, Building Slab Top: 68.82 M.
05/TKOF	BUILDING	230437.1N 0723811.1E	206 FT	NIL	Building, Sintex Tank Top: 62.90 M, Building Top: 57.40 M.
In circling area and at AD	BUILDING	230435.9N 0723811.8E	215 FT	NIL	Building, Staircase Room Top: 65.58 M, Building Top: 64.14 M
In circling area and at AD	BUILDING	230438.0N 0723811.6E	199 FT	NIL	Building Top: 60.6 M
In circling area and at AD	BUILDING	230500.9N 0723839.2E	237 FT	NIL	Building, Water Tank Top: 72.37 M, Building Top: 69.89 M.
In circling area and at AD	BUILDING	230501.7N 0723840.1E	238 FT	NIL	Building, Water Tank Top: 72.45 M, Building Top: 69.83 M.
In circling area and at AD	TREE	230347.9N 0723656.2E	251 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	230510.9N 0723850.8E	258 FT	NIL	TREE
In circling area and at AD	TREE	230510.8N 0723850.2E	248 FT	NIL	TREE
In circling area and at AD	BUILDING	230604.7N 0723635.6E	350 FT	NIL	Sharan Sapphire Apartment, Sintex Tank Top: 106.61 M, Staircase Top: 104.15 M, Building Top: 102.27 M
In circling area and at AD	BUILDING	230605.3N 0723641.6E	352 FT	NIL	Building (U/C) Top
In circling area and at AD	BUILDING	230604.0N 0723643.2E	361 FT	LGTD	Building (U/C), Sintex Tank Top: 110.13 M, Building Top: 108.24 M.

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	230604.2N 0723652.0E	342 FT	NIL	Sampad Woods Block-C Building, Water Tank Top: 104.14 M, Building Top: 100.90 M.
In circling area and at AD	BUILDING	230606.4N 0723654.5E	342 FT	NIL	Sampad Woods Building, Antenna Top: 104.29 M, Water Tank Top: 103.94 M, Building Top: 101.82 M.
In circling area and at AD	BUILDING	230606.4N 0723652.4E	348 FT	NIL	Sampad Woods Building, Water Tank Top: 106.22 M, Building Top: 104.31 M.
In circling area and at AD	BUILDING	230605.5N 0723654.2E	341 FT	NIL	Sampad Woods Building, Water Tank Top: 104.06 M, Building Top: 102.40
In circling area and at AD	BUILDING	230548.4N 0723944.6E	337 FT	LGTD	Rashmi Antelia, Building Top
In circling area and at AD	BUILDING	230650.8N 0723929.8E	345 FT	LGTD	Radhe Skyline-Ii, Water Tank Top: 105.04 M, Building Top: 104.50 M.
In circling area and at AD	BUILDING	230551.8N 0724106.7E	371 FT	NIL	Naroda Icon, Water Tank Top: 112.96 M, Building Top: 105.70 M.
In circling area and at AD	TRANSMISSION LINE	230654.1N 0723923.3E	377 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230626.0N 0724048.5E	350 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230620.9N 0724053.6E	357 FT	NIL	H.T. Tower
In circling area and at AD	BUILDING	230424.6N 0723541.1E	368 FT	LGTD	Torrent Power Limited Factory, Chimney Top: 112.16 M, Chimney Top: 111.26 M, Factory Shed Top: 106.71 M.
In circling area and at AD	BUILDING	230412.0N 0723537.4E	362 FT	LGTD	Torrent Power Limited Factory, Chimney Top: 110.33 M, Factory Shed Top: 107.34 M.
In circling area and at AD	BUILDING	230412.3N 0723536.8E	347 FT	LGTD	Torrent Power Limited Factory, Building Top

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	230413.9N 0723536.0E	354 FT	LGTD	Torrent Power Limited Factory, Chimney Top: 107.89 M, Factory Shed Top: 106.78 M.
In circling area and at AD	BUILDING	230414.0N 0723535.5E	352 FT	LGTD	Torrent Power Limited Factory, Building Top
In circling area and at AD	BUILDING	230414.4N 0723526.7E	364 FT	LGTD	Torrent Power Limited Factory, Chimney Top: 110.87 M, Factory Shed Top: 105.67 M.
In circling area and at AD	BUILDING	230519.1N 0724102.2E	369 FT	LGTD	Murlidhar Heights (U/C), Water Tank Top: 112.71 M, Building Top: 109.99 M.
In circling area and at AD	BUILDING	230518.7N 0724101.4E	370 FT	LGTD	Murlidhar Heights (U/C), Water Tank Top: 112.68 M, Building Top: 110.01 M.
In circling area and at AD	BUILDING	230518.0N 0724059.9E	370 FT	LGTD	Murlidhar Heights (U/C), Building Top
In circling area and at AD	BUILDING	230520.8N 0724101.3E	370 FT	LGTD	Murlidhar Heights, Water Tank Top: 112.71 M, Building Top: 109.90 M.
In circling area and at AD	STACK	230422.0N 0723539.5E	469 FT	LGTD	Chimney
In circling area and at AD	STACK	230413.5N 0723539.8E	475 FT	LGTD	Chimney
In circling area and at AD	STACK	230415.0N 0723537.9E	476 FT	LGTD	Chimney
In circling area and at AD	STACK	230412.9N 0723524.3E	466 FT	LGTD	Chimney
In circling area and at AD	TRANSMISSION LINE	230656.3N 0723829.0E	387 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230702.9N 0723845.1E	379 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230659.9N 0723841.9E	379 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230656.5N 0723838.3E	381 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230646.5N 0723844.5E	435 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230645.2N 0723823.7E	381 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230653.1N 0723824.3E	382 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230647.3N 0723907.8E	431 FT	NIL	H.T. Tower

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	TRANSMISSION LINE	230657.8N 0723914.9E	366 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230704.5N 0723914.5E	371 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230706.0N 0723907.7E	356 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230705.4N 0723903.2E	342 FT	NIL	H.T. Tower
In circling area and at AD	TRANSMISSION LINE	230704.0N 0723853.2E	345 FT	NIL	H.T. Tower
In circling area and at AD	BUILDING	230429.7N 0723954.7E	341 FT	LGTD	Sahitya Icon (U/C), Building Top
In circling area and at AD	BUILDING	230429.1N 0723955.5E	351 FT	LGTD	Sahitya Icon (U/C), Water Tank Top: 107.03 M, Building Top: 104.02 M.
In circling area and at AD	FENCE	230519.2N 0723855.8E	205 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	OTHER	230526.6N 0723848.0E	205 FT	NIL	Mobile Road Traffic (Road Elev – 57.43 M + 5 M Mobile Traffic)
In circling area and at AD	NAVAID	230350.5N 0723730.4E	279 FT	LGTD	SMGCS Radar
In circling area and at AD	NAVAID	230414.9N 0723752.0E	262 FT	LGTD	L/C Rod on MSSR (MSSR Top: 74.3 M.)
In circling area and at AD	NAVAID	230513.7N 0723835.9E	242 FT	LGTD	GP Main Antenna (Lighted Top: 72.4 M)
In circling area and at AD	NAVAID	230513.8N 0723835.8E	212 FT	LGTD	GP DME Antenna
In circling area and at AD	NAVAID	230515.5N 0723837.9E	207 FT	LGTD	GP Monitor Antenna
In circling area and at AD	BUILDING	230534.2N 0723553.9E	347 FT	LGTD	Narendra Modi Stadium
In circling area and at AD	TREE	230508.3N 0723848.9E	257 FT	NIL	GROUP OF TREES
In circling area and at AD	TRANSMISSION LINE	230623.2N 0724046.9E	349 FT	NIL	H.T. Tower
In circling area and at AD	BUILDING	230513.7N 0724045.4E	351 FT	NIL	Building Pratishtha Icon Block B
In circling area and at AD	BUILDING	230514.1N 0724046.1E	351 FT	NIL	Building Pratishtha Icon Block C
In circling area and at AD	BUILDING	230512.7N 0724045.3E	351 FT	NIL	Building Pratishtha Icon Block A
In circling area and at AD	BUILDING	230513.3N 0724046.6E	351 FT	NIL	Building Pratishtha Icon Block D
In circling area and at AD	BUILDING	230621.8N 0723646.1E	368 FT	NIL	U/C Building Block B

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	230621.7N 0723645.1E	369 FT	NIL	U/C Building Block C
In circling area and at AD	BUILDING	230503.3N 0724106.3E	376 FT	NIL	U/C Building Ganesh Heritage, Block A
In circling area and at AD	BUILDING	230502.9N 0724107.2E	375 FT	NIL	U/C Building Ganesh Heritage, Block B+C
In circling area and at AD	BUILDING	230503.4N 0724109.2E	375 FT	NIL	U/C Building Ganesh Heritage, Block. D
In circling area and at AD	BUILDING	230435.9N 0724054.8E	373 FT	NIL	U/C Building Block F+G
In circling area and at AD	BUILDING	230435.6N 0724056.3E	373 FT	NIL	U/C Building Block I+H
In circling area and at AD	BUILDING	230433.8N 0724055.7E	373 FT	NIL	U/C Building Block J+K
In circling area and at AD	BUILDING	230510.8N 0724049.3E	352 FT	NIL	U/C Building Dev Aashish Z Plus Block B+C
In circling area and at AD	BUILDING	230511.9N 0724051.5E	352 FT	NIL	U/C Building Dev Aashish Z Plus Block A
In circling area and at AD	BUILDING	230511.8N 0724045.7E	352 FT	NIL	U/C Building Nand Vatika 2, Hoarding 106.71 M, Building Top: 105.176 M
In circling area and at AD	BUILDING	230512.8N 0724047.7E	352 FT	NIL	U/C Building Nand Vatika 2, Building Top: 105.13 M
In circling area and at AD	BUILDING	230511.5N 0724048.9E	355 FT	NIL	U/C Building Himshila Sky. Block B+C+D, 106.79 M
In circling area and at AD	BUILDING	230513.0N 0724051.0E	355 FT	NIL	U/C Building Himshila Sky. Block A 105.53 M
In circling area and at AD	BUILDING	230547.3N 0723949.6E	344 FT	NIL	U/C Building SONET HEIGHTS, Block A
In circling area and at AD	BUILDING	230547.8N 0723950.4E	344 FT	NIL	U/C Building SONET HEIGHTS, Block B+C
In circling area and at AD	BUILDING	230630.9N 0723921.2E	354 FT	NIL	U/C Building, Ornet Height, Block A
In circling area and at AD	BUILDING	230631.4N 0723919.5E	355 FT	NIL	U/C Building, Ornet Height, Block B
In circling area and at AD	BUILDING	230644.8N 0723928.2E	348 FT	NIL	Mapple Heights, Block B, Pipe on Building: 106.16 M, Building Top: -105.92 M

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	230646.5N 0723928.8E	348 FT	NIL	Mapple Heights, Block A, Pipe on Building: 106.15 M, Building Top 105.859
In circling area and at AD	BUILDING	230644.5N 0723929.4E	348 FT	NIL	Mapple Heights, Block A, Pipe on Building 106.18, Building Top: 105.79 M
In circling area and at AD	BUILDING	230503.5N 0724047.2E	361 FT	NIL	U/C Building, Dev Ashish Pride & Plaza Block B+C, Building Top: 106.40 M
In circling area and at AD	BUILDING	230502.8N 0724046.4E	371 FT	NIL	U/C Building, Dev Ashish Pride & Plaza Block D, Building Top: 106.50 M
In circling area and at AD	BUILDING	230509.3N 0724108.9E	380 FT	NIL	U/C Building Ganesh Glorious Block D
In circling area and at AD	BUILDING	230509.3N 0724109.9E	380 FT	NIL	U/C Building Ganesh Glorious Block B+C
In circling area and at AD	BUILDING	230509.5N 0724112.0E	381 FT	NIL	U/C Building Ganesh Glorious Block A
In circling area and at AD	BUILDING	230514.8N 0724052.0E	359 FT	NIL	U/C Building, Signature Sky, Block A+B
In circling area and at AD	BUILDING	230514.9N 0724053.1E	359 FT	NIL	U/C Building, Signature Sky, Block C+D
In circling area and at AD	BUILDING	230513.1N 0724048.5E	368 FT	NIL	U/C Building Shivanta Sky Block C+D
In circling area and at AD	BUILDING	230513.8N 0724049.9E	368 FT	NIL	U/C Building Shivanta Sky Block B
In circling area and at AD	BUILDING	230514.1N 0724050.6E	368 FT	NIL	U/C Building Shivanta Sky Block A
In circling area and at AD	BUILDING	230459.2N 0724106.6E	354 FT	NIL	U/C Building Naroda SKY Block A+B
In circling area and at AD	BUILDING	230458.8N 0724107.8E	356 FT	NIL	U/C Building Naroda SKY Block C+D
In circling area and at AD	BUILDING	230459.4N 0724109.6E	359 FT	NIL	U/C Building Naroda SKY Block E+F
In circling area and at AD	BUILDING	230505.3N 0724110.1E	388 FT	NIL	U/C Building Fortune Empire Block C

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	230504.9N 0724111.1E	378 FT	NIL	U/C Building Fortune Empire Block B
In circling area and at AD	BUILDING	230603.6N 0723642.0E	354 FT	NIL	Building Kiara Monolith, Pipe on The Monolith: 109.13 M, Water Tank Top: 108.07M
In circling area and at AD	BUILDING	230715.3N 0723928.8E	380 FT	NIL	U/C Building Keshvam Heights Block A+B
In circling area and at AD	BUILDING	230715.4N 0723927.7E	355 FT	NIL	U/C Building Keshvam Heights Block C
In circling area and at AD	BUILDING	230717.4N 0723928.6E	382 FT	NIL	U/C Building Keshvam Heights Block A1+B1
In circling area and at AD	BUILDING	230717.4N 0723927.8E	376 FT	NIL	U/C Building Keshvam Heights Block C1
In circling area and at AD	BUILDING	230554.7N 0724100.7E	368 FT	NIL	U/c Building Murlidhar Glory Block G-H
In circling area and at AD	BUILDING	230554.0N 0724101.6E	368 FT	NIL	U/c Building Murlidhar Glory Block E-F
In circling area and at AD	BUILDING	230553.2N 0724101.8E	368 FT	NIL	U/c Building Murlidhar Glory Block D
In circling area and at AD	BUILDING	230554.3N 0724103.6E	368 FT	NIL	U/c Building Murlidhar Glory Block C
In circling area and at AD	BUILDING	230501.7N 0724046.7E	394 FT	NIL	U/c Building Dev Aashish Pride Plus
In circling area and at AD	BUILDING	230602.2N 0723701.6E	344 FT	NIL	U/c Building Kiara Signature, Block C
In circling area and at AD	BUILDING	230601.3N 0723701.5E	343 FT	NIL	U/c Building Kiara Signature, Block A
In circling area and at AD	BUILDING	230602.1N 0723700.3E	344 FT	NIL	U/c Building Kiara Signature, Block B
In circling area and at AD	BUILDING	230635.1N 0723805.8E	365 FT	NIL	U/c Building Block A
In circling area and at AD	BUILDING	230635.2N 0723804.2E	366 FT	NIL	U/c Building Block B
In circling area and at AD	BUILDING	230630.5N 0723754.3E	359 FT	NIL	Building German Green City, Block A Solar on Building: 104.73 M
In circling area and at AD	BUILDING	230630.6N 0723753.2E	359 FT	NIL	Building German Green City, Block B Water Tank on Building: 105.56 M

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	230631.7N 0723753.3E	359 FT	NIL	Building German Green City, Block C+D Shed on Building: 104.83 M
In circling area and at AD	BUILDING	230508.2N 0724042.3E	376 FT	NIL	U/c Building White Crest Building Top: 104.93 M
In circling area and at AD	BUILDING	230605.7N 0723651.5E	351 FT	NIL	Building Polaris Celestia B, Solar on Building, Water Tank Top: 104.031 M, Building Top: 103.384 M
In circling area and at AD	BUILDING	230605.6N 0723649.9E	352 FT	NIL	Building Polaris Celestia Block A, Solar on Building, Water Tank Top: 103.920 M, Building Top: 103.313 M
In circling area and at AD	BUILDING	230432.5N 0724028.7E	374 FT	NIL	U/c Building Kahaan Block A
In circling area and at AD	BUILDING	230637.2N 0723923.1E	358 FT	NIL	U/c Building, Nakshatra Skyview, Block A+B
In circling area and at AD	BUILDING	230637.5N 0723922.1E	359 FT	NIL	U/c Building, Nakshatra Skyview, Block C
In circling area and at AD	BUILDING	230637.8N 0723921.0E	360 FT	NIL	U/c Building, Nakshatra Skyview, Block D
In circling area and at AD	BUILDING	230639.1N 0723921.4E	361 FT	NIL	U/c Building, Nakshatra Skyview, Block E
In circling area and at AD	BUILDING	230502.0N 0724109.6E	394 FT	NIL	U/c Building Krishna Heritage Block D
In circling area and at AD	BUILDING	230607.6N 0723631.4E	363 FT	NIL	U/c Building Block C
In circling area and at AD	BUILDING	230611.4N 0723631.5E	364 FT	NIL	U/c Building Block A+B
In circling area and at AD	BUILDING	230607.4N 0723629.7E	363 FT	NIL	U/c Building Block D
In circling area and at AD	BUILDING	230552.9N 0724103.6E	382 FT	NIL	U/c Building Murlidhar Glory Block A+B
In circling area and at AD	BUILDING	230507.4N 0723841.5E	239 FT	LGTD	GLIDEPATH 23 SOUTH SIDE

VAAH AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	Ahmedabad
2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	H-24

3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	Ahmedabad 9 and 24 HR
4	Availability of the trend forecast for the aerodrome and interval of issuance	Trend METAR/SPECI/Special reports with Trend 30 Min
5	Information on how briefing and/or consultation is provided	Provided with en-route forecast
6	Types of flight documentation supplied and language(s) used in flight documentation	Tabular Form (English)
7	Charts and other information displayed or available for briefing or consultation	S, U85, U70, U50 U30 U20.
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	Telex, Telefax, Satellite display workstation. DCWIS and Forward Scatter Meter AVBL at RWY 05 End. Following Met parameters are AVBL at RWY 05 End: 1.Pressure (QNH) 2.Dry Bulb Temperature 3.Dew Point Temperature 4.Wind Direction 5.Wind Speed 6.RVR
9	The air traffic services unit(s) provided with meteorological information	Ahmedabad ATS
10	Additional information, e.g. concerning any limitation of service.	i. Advance notice of 2 Hrs required for domestic routes and 12 Hrs notice required for INTL routes. ii. New Digital Current Weather Indicating System and Forward Scatterometer operational at RWY 23 END, located at a distance of 290M from RWY 23 Threshold and 120M from Centreline. The following MET parameters are available at RWY 23 END: Pressure (QNH), Dry Bulb, Temperature, Dew Point Temperature Wind Direction,Wind Speed and RVR.

VAAH AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCR) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	2	3	4	5
05	44.67 DEG	3505 x 45 M	862/F/A/W/T Asphalt	THR: 230357.45N 0723720.66E RWY END: 230518.48N 0723847.25E
23	224.67 DEG	3505 x 45 M	862/F/A/W/T Asphalt	THR: 230518.48N 0723847.25E RWY END: 230357.45N 0723720.66E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 180.0FT TDZ: 187.0FT	0.07%	NIL	NIL	3625 x 280 M
THR: 189.0FT TDZ: 189.0FT	-0.07%	NIL	NIL	3625 x 280 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
130M x 90M		NIL	i. RWY Strip of 140M width available on either side of RWY 05/23 CL ii. Gravity Model: EGM 08
240M x 90M		NIL	i. RWY Strip of 140M width available on either side of RWY 05/23 CL ii. Gravity Model: EGM 08

VAAH AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
05	3505	3505	3505	3505	1:50
23	3505	3505	3505	3505	1:50

VAAH AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	2	3	4	5
05	SALS 420 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (66.58FT)	
23	CAT I 750 M ALS LIH	Green	PAPI LEFT/3.00 DEG MEHT (68.20FT)	

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
	3505 M 60 M LIH	Red		NIL
	3505 M 60 M LIH	Red		NIL

VAAH AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	At 230440.59N 0723750.85E On top of the ARFF building – Flashing White and Green lights alternatively- 30FPM Operation Hours: During Night and Low visibility
		IBN	NIL
2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	Runway 05/23: Illuminated Landing T provided northeast of taxiway D.
		Anemometer	RWY23 at 762M from beginning.
3	Taxiway edge and taxiway centre line lights;	Edge	TWY Edge lights Available on A, B, C, D, F, G, H, K, M, P, R4, R, R1, R3, R5
		Centre Line	Available on RET R5
4	Secondary power supply including switch-over time;	Secondary power supply to all lighting at AD. Switch-over time :15 SEC.	
5	Remarks	1. Apron edge and flood lights available. 2.WDI provided at both ends of Runway north of aiming point of Runway 23 & South of Runway 05 and northeast of Taxiway-F.	

VAAH AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	Not established
2	TLOF and/or FATO area elevation:	Not established
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not established
4	True bearings of FATO;	Not established
5	Declared distances available	Not established
6	Approach and FATO lighting;	Not established
7	Remarks	Not established

VAAH AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Circular area centered on 230416N 0723735E within a 15NM radius.
2	Vertical limits	FL 65
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Ahmedabad Tower / Ahmedabad Approach, English
5	Transition altitude	4000 FT
6	Hours of applicability	H24
7	Remarks	1. Class D airspace below 5500 ft AMSL. 2. Class C airspace at or above 5500 ft AMSL.

VAAH AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
1	2	3	4
TAR	Ahmedabad Radar	119.800 MHZ	
APP	Ahmedabad Approach	119.800 MHZ	
APP	Ahmedabad Approach	128.375 MHZ	
TWR	Ahmedabad Tower	118.100 MHZ	
TWR	Ahmedabad Tower	119.600 MHZ	
ATIS	Ahmedabad Information	126.800 MHZ	
ACC	Ahmedabad Control	119.350 MHZ	
ACC	Ahmedabad Control	123.750 MHZ	
ACC	Ahmedabad Control	134.200 MHZ	
ALRS	Emergency Frequency	121.500 MHZ	
RADAR	Ahmedabad Radar	123.750 MHZ	
RADAR	Ahmedabad Radar	134.200 MHZ	
SMC	Ahmedabad Ground	121.675 MHZ	

Logon address, as appropriate	Hours of operation	Remarks
5	6	7
	H24	NIL
	H24	Main
	H24	Backup
	H24	Tower Cum SMC Backup
	H24	Main
	H24	NIL
	H24	SDBY FREQUENCY FOR AHMEDABAD ACC (LOWER) AND AHMEDABAD ACC (UPPER)
	H24	Primary Frequency . Primary Frequency RCAG installed at Udaipur and Bhuj
	H24	Secondary Frequency
	H24	NIL
	H24	1. Primary Frequency. RCAG installed at Udaipur and Bhuj. 2. Surveillance based Area Control Service will be bifurcated into Upper and Lower Area Control Sectors as per Traffice requirements, Lower Area Control Freq: 134.2 MHZ Upper Area Control Freq: 123.75 MHZ. Radio Callsign for both sectors is Ahmedabad Radar.
	H24	1. Secondary Frequency. RCAG installed at Indore 2. Surveillance based Area Control Service will be bifurcated into Upper and Lower Area Control Sectors as per Traffice requirements, Lower Area Control Freq: 134.2 MHZ Upper Area Control Freq: 123.75 MHZ. Radio Callsign for both sectors is Ahmedabad Radar.
	H24	Main

VAAH AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 23 CAT I	IAHD	110.300 MHz	H24
GP 23		335.000 MHz	H24
DME ILS 23	IAHD	CH40X	H24
DVOR/DME	AAE	113.100 MHz CH78X	H24

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
230349.3N 0723712.0E			
230507.4N 0723841.5E			3 DEG
230507.3N 0723841.7E	201 FT		Collocated with GP23. Elevation in EGM08.
230407.3N 0723743.5E	216 FT		1. Vertical datum: EGM08

VAAH AD 2.20 LOCAL AERODROME REGULATIONS

I. Start up and Taxi procedure at Apron 3:

S.No.	Aircraft Stand	Start up and Taxi Procedure
1	25 to 30	i) Aircraft taxing-in for stand 25 to 30 shall follow TWY L5 and TXL L6. ii) Aircraft taxing out from stand 25 to 30 shall follow TXL L7 and TWY L5.
2	25A to 30A	i) Aircraft taxing-in for stand 25A to 30A shall follow TWY L5, and TXL L8. ii) Aircraft taxing out from stand 25A to 30A shall follow TXL L7, TWY L5.

Restriction of ACFT movement on TWY P

Taxiway	Code of Aircraft at Holding point	Code of Aircraft permitted to taxi behind on TWY "P"
A or B	E	None
A or B	D (Max length 55M)	Up to C

A or B	Up to C	All Aircraft up to E
C	E	Up to C
C	Upto D (Max Length 55M)	All Aircraft up to E
D	E	A/B
D	D	Up to C
D	C	All Aircraft up to D
D	A/B	All Aircraft up to E
R4	D/E	None
R4	Up to C	Up to Code C

Note:

a) ATC may ask code C, D or E aircraft taxiing on taxiway P to hold on intermediate holding position “P1” to provide rapid access for fire-fighting & rescue vehicles to runway. Marking and signage board for “P1” provided for easy identification.

b) To reduce RWY occupancy time, while backtracking on RWY all ACFT shall maintain speed of 25 KT or more DRG dry RWY condition.

c) REQ for turn pad/full length or TWY D/R4 INT DEP for RWY 23, shall be informed to ATC at the time of push back/start up.

II. Pushback Procedure (for Aircraft Stands 31 to 43):

Pushback and taxi Procedure for stand 31A, 31L, 31, 31R, 32L, 32, 32R, 33, 34L,34, 34R, 35, 36L, 36, 36R, 37, 38, 39, 40, 41, 42 and 43:

i. All stands in the apron 2 are power-in/pushback.

ii. Stand no. 31, 32, 34 and 36 are MARS stands.

iii. When code D/E parked on stand 33 no parking on stand 32R, however, when a Code C aircraft is parked on stand 33, parking of another code C aircraft is permitted on stand 32R.

iv. Stand 35 is independent up to code C.

v. Stand 37 to 43 are compatible for code A/B/C.

vi. The pushback operator shall ensure that the aircraft is aligned on the taxi lane and nose wheel of aircraft should be abeam stand unless otherwise advised by ATC.

vii. Ensuring correct push back on taxilane L10 as per instructions received from ATC is the responsibility of Airline operator/ GHA.

III. <u>Pushback and Taxi Out Procedure via Taxiway G</u> (for Aircraft Stands 31 to 43):

Aircraft Stand	Pushback Procedure	Pushback not permitted	Docking not permitted
31A, 31L and 31R	Pushback facing southwest on Taxilane L10 and deep push abeam stand 32L.	Stand 31L, 31A, 31R, 31, 32L, 32, 32R, 33, 37 and 38.	Stand 31L, 31A, 31R, 31, 32L, 32, 32R, 33, 37, 38. **(Note: Stand 31A, and 31L, docking may be permitted post aircraft reported abeam stand 32L)**.
31	Pushback facing southwest on taxilane L10 and deep push abeam stand 32L.	Stand 31A, 31L, 31R, 32L, 32, 32R, 33, 34L, 34, 34R, 37, 38, 39, 40 and 41.	Stand 31A, 31L,31R, 32L, 32, 32R, 33, 37, 38 and 39. **(Note: Stand 31A, and Airports Authority of India AIP Supplement 72/2024 Page 5 of 11 31L, docking may be permitted post aircraft reported abeam stand 32L).**
32L	Pushback facing southwest on taxilane L10 abeam stand 32L.	Stand 31A, 31L, 31, 31R, 32, 32R, 33, 37 and 38.	Stand 31, 31R, 32, 32R, 33, 37 and 38.
32	Pushback facing southwest on taxilane L10 abeam stand 32.	Stand 31A, 31L, 31, 31R, 32L, 32R, 33, 34L, 34, 34R, 35, 37, 38, 39,40,41 and 42.	Stand 31, 31R, 32L,32R, 33, 34L, 34, 37, 38, 39 and 40.
32R	Pushback facing southwest on taxilane L10 abeam stand 32R.	Stand 31A, 31L, 31, 31R, 32L, 32, 33, 37, 38, and 39.	Stand 32L, 32, 33, 37, 38 and 39.
33	Pushback facing southwest on taxilane L10 abeam stand 33.	Stand 31A, 31L, 31, 31R, 32R, 32, 34L, 34, 34R, 35, 36L, 36, 36R, 37, 38, 39, 40, 41, 42 and 43.	Stand 32, 32R, 34L, 34, 34R, 35, 38, 39, 40, 41 and 42.
34L	Pushback facing southwest on taxilane L10 abeam stand 34L.	Stand 31, 32, 33, 34, 34R, 35, 36, 38, 39, 40, 41 and 42.	Stand 33, 34, 34R, 35 39, 40, 41 and 42.
34	Pushback facing southwest on taxilane L10 abeam stand 34.	Stand 31, 32, 33, 34L, 34R, 35, 36L, 36, 36R, 38, 39, 40, 41, 42 and 43.	Stand 33, 34L, 34R, 35, 36L, 36, 39, 40, 41, 42 and 43.
34R	Pushback facing southwest on taxilane L10 abeam stand 34R.	Stand 31, 32, 33, 34L, 34, 35, 36L, 36, 36R, 39, 40, 41, 42 and 43	Stand 34L, 34, 35, 36L, 40, 41, 42 and 43.
35	Pushback facing southwest on taxilane L10 abeam stand 35.	Stand 32, 33, 34L, 34, 34R, 36L, 36, 36R, 40, 41, 42 and 43.	Stand 34R, 36L, 36, 36R, 41, 42, and 43.

36L	Pushback facing southwest on taxilane L10 abeam stand 36L.	Stand 33, 34, 34R, 35, 36, 36R, 41, 42 and 43.	Stand 35, 36, 36R, 42 and 43.
36	Pushback on taxiway H.	Stand 35, 36L, 36R, 42 and 43.	Stand 36L, 36R, and 43.
36R	Pushback on taxiway H.	Stand 35, 36L, 36, 42 and 43.	Stand 36L, 36, and 43.
37	Pushback facing Southwest on taxilane L10 abeam stand 37.	Stand 31A, 31L, 31, 31R, 32L, 32, 32R, 33, 38.	Stand 32L, 32, 32R, 33, 38 and 39.
38	Pushback facing Southwest on taxilane L10 abeam stand 38.	Stand 31A, 31L, 31, 31R, 32L, 32, 32R, 33, 34L, 34, 37, 39 and 40.	Stand 32, 32R, 33, 34L, 34, 37, 39 and 40.
39	Pushback facing Southwest on taxilane L10 abeam stand 39.	Stand 31, 32, 32R, 33, 34L, 34, 34R, 37, 38 40 and 41.	Stand 33, 34L, 34, 34R, 38, 40 and 41.
40	Pushback facing Southwest on taxilane L10 abeam stand 40.	Stand 31, 32, 33, 34L 34, 34R, 35, 38, 39, 41 and 42.	Stand 33, 34L, 34, 34R, 35, 39, 41and 42.
41	Pushback facing Southwest on taxilane L10 abeam stand 41.	Stand 31, 32, 33, 34L 34, 34R, 35, 36L, 39, 40 42 and 43.	Stand 34L, 34, 34R, 35, 36L, 40, 42 and 43.
42	Pushback facing Southwest on taxilane L10 abeam stand 42.	Stand 31, 32, 33, 34L, 34, 34R, 35, 36L, 36, 36R, 40, 41 and 43.	Stand 34R, 35, 36L, 36 36R, 41 and 43.
43	Pushback facing Southwest on taxilane L10 abeam stand 43.	Stand 33, 34, 34R, 35, 36L, 36, 36R, 41 and 42.	Stand 35, 36L, 36, 36R, and 42.

IV. <u>Pushback and Taxi Out Procedure via Taxiway H</u> (for Aircraft Stands 31 to 43):

Aircraft Stand	Pushback Procedure	Pushback not permitted	Docking not permitted
31A	Pushback on Taxiway G to taxi via taxilane L10 - taxiway H.	Stand 31L, 31R, 31, 32L, 32, 32R, 33, 34, 37 and 38.	Stand 31L, 31R, 31, 32L, 32, and 37.
31L and 31R	Pushback facing Northeast on taxilane L10 and pull abeam stand 32L.	Stand 31A, 31L, 31R, 31, 32L, 32, 32R, 33, 34, 37, and 38.	Stand 31A, 31L, 31R, 31, 32L, 32, 32R and 37.
31	Pushback on taxiway G for startup and taxi out via taxilane L10 to taxiway H.	Stand 31A, 31L, 31R, 32L, 32, 32R, 33, and 37.	Stand 31A, 31L, 31R, 32L, 32, and 37.
32L	Pushback facing Northeast on taxilane L10 abeam stand 32L.	Stand 31A, 31L, 31, 31R, 32, 32R, 33, 34, 37 and 38.	Stand 31A, 31L, 31, 31R, 32, 32R and 37.
32	Pushback facing Northeast on taxilane L10 and pull abeam stand 33 for startup and taxi out via taxiway H.	Stand 31A, 31L, 31, 31R, 32L, 32R, 33, 34L, 34, 36, 37, 38, 39 and 40.	Stand 31A, 31L, 31, 31R, 32L, 32R, 33, 34L, 34, 37, 38, 39, and 40. **(Note: Stand 31A, 31L, 31 and 31R docking may be permitted post aircraft reported abeam stand 33).**
32R	Pushback facing Northeast on taxilane L10 abeam stand 32R.	Stand 31L, 31, 31R, 32L, 32, 33, 34, 37, 38 and 39.	Stand 31, 31R, 32L, 32, 33, 37 and 38.
33	Pushback facing Northeast on taxilane L10 abeam stand 33.	Stand 31A, 31L 31, 31R, 32R, 32, 32L, 34L, 34, 36, 37, 38, 39 and 40.	Stand 32L, 32, 32R, 34L, 34, 37, 38, 39, and 40.
34L	Pushback facing Northeast on taxilane L10 abeam stand 34L.	Stand 32, 33, 34, 34R, 35, 36, 38, 39, 40, 41 and 42.	Stand 33, 34, 34R, 38, 39, 40 and 41.
34	Pushback facing Northeast on taxilane L10 abeam stand 34.	Stand 31L, 31R, 32L, 32, 32R, 33, 34L, 34R, 35, 36, 37, 38, 39, 40, 41 and 42.	Stand 32L, 32, 32R, 33, 34L, 34R, 37, 38, 39, 40 and 41.
34R	Pushback facing Northeast on taxilane L10 abeam stand 34R.	Stand 34L, 34, 35, 36, 36L, 39, 40, 41, 42 and 43.	Stand 34L, 34, 35, 39, 40, 41 and 42.
35	Pushback facing Northeast on taxilane L10 abeam stand 35.	Stand 34L, 34, 34R, 36L, 36, 36R, 40, 41, 42 and 43.	Stand 34L, 34, 34R, 36L, 36, 40, 41, 42 and 43.
36L	Pushback facing Northeast on taxilane L10 abeam stand 36L.	Stand 34R, 35, 36, 36R, 41, 42 and 43.	Stand 34R, 35, 36, 36R, 41, 42 and 43.

36	Pushback facing Northeast on taxilane L10 abeam stand 36.	Stand 32, 33, 34L, 34, 34R, 35, 36L, 36R, 39, 40, 41, 42 and 43.	Stand 34L, 34, 34R, 35, 36L, 36R, 40, 41, 42, and 43.
36R	Pushback facing northeast on taxilane L10 abeam stand 36R.	Stand 35, 36L, 36, 42 and 43.	Stand 35, 36L, 36, 42 and 43.
37	Pushback facing Northeast on taxilane L10 abeam stand 37.	Stand 31A, 31L, 31, 31R, 32L, 32, 32R, 33, 34, 38 and 39.	Stand 31A, 31L, 31, 31R, 32L, 32, 32R, 33 and 38.
38	Pushback facing Northeast on taxilane L10 abeam stand 38.	Stand 31L, 31R, 32L, 32, 32R, 33, 34, 34L, 36, 37, 39 and 40.	Stand 32L, 32, 32R, 33, 37 and 39.
39	Pushback facing Northeast on taxilane L10 abeam stand 39.	Stand 32, 32R, 33, 34L, 34, 34R, 36, 37, 38, 40 and 41.	Stand 32, 32R, 33, 34L, 34, 37, 38, and 40.
40	Pushback facing Northeast on taxilane L10 abeam stand 40.	Stand 33, 34L 34, 34R, 35, 36, 38, 39, 41 and 42.	Stand 33, 34L, 34, 34R, 38, 39 and 41.
41	Pushback facing Northeast on taxilane L10 abeam stand 41.	Stand 34L, 34, 34R, 35, 36L, 36, 39, 40, 42 and 43.	Stand 34L, 34, 34R, 35, 39, 40 and 42.
42	Pushback facing Northeast on taxilane L10 abeam stand 42.	Stand 34L, 34, 34R, 35, 36L, 36, 36R, 40, 41 and 43.	Stand 34L, 34, 34R, 35, 36L, 36, 40, 41 and 43.
43	Pushback facing Northeast on taxilane L10 abeam stand 43.	Stand 34R, 35, 36L, 36, 36R, 41 and 42.	Stand 34R, 35, 36L, 36, 36R, 41 and 42.

V. <u>For Aircraft Stands 70 to 94:</u>

<u>A. Pushback and Start Up Procedure</u>:

70	Code D/E aircraft pushback from stand 70 shall deep push and pull ahead abeam stand 72 for startup on taxiway L9 facing southeast.	Stand 70L, 70R, 71L, 71, 71R, 72	Stand 70L, 70R, 71L, 71, 71R, 72, 73
70L	Code C Aircraft Pushback from stand 70L to pull abeam stand 71L on Taxiway L4 facing southeast for startup.	Stand 70, 70R, 71L, 71, 71R, 72	Stand 70, 70R, 71L, 71, 71R
70R	Code C Aircraft Pushback from stand 70R to pull abeam stand 71L on Taxiway L4 facing southeast for startup.	Stand 70L, 70, 71L, 71, 71R, 72	Stand 70L, 70, 71L, 71, 71R, 72, 73
71	Code D aircraft Pushback from stand 71 shall be on taxiway L9 and pull abeam stand 72 for startup	Stand 70L, 70, 70R, 71L, 71R, 72, 73, 74, 75	Stand 70L, 70, 70R, 71L, 71R, 72, 73, 74
71L	Code C Aircraft Pushback and startup abeam stand 71L on Taxiway L4 facing southeast.	Stand 70L, 70, 70R, 71R,71, 72, 73	Stand 70L, 70, 70R, 71R, 71, 72
71R	Code C Aircraft Pushback and startup abeam stand 71R on Taxiway L4 facing southeast.	Stand 70L, 70, 70R, 71L, 71, 72, 73, 74	Stand 70L, 70, 70R, 71L, 71, 72, 73
72 to 83	All Code C Aircraft Pushback and startup abeam respective stands on Taxiway L4 facing southeast.	Minimum stand separation required for simultaneous push-back and startup for Code C aircraft on Apron 1 is 03 stands. **<u>Exception:</u>** a) If required, ATC may instruct to pushback/pull forward from stands 80 and 81 to abeam stand 79/82 on taxiway L4 to keep clear link taxiways T1 and T3 crossing. b) ATC may handle traffic on contact stands from 77 to 80 tactically for optimum utilization of aerobridges.	Minimum 02 stand separation shall be required to maintain between pushback aircraft and docking aircraft on Apron 1.

84 to 94	All Code C Aircraft Pushback and startup abeam respective stands on Taxiway L3 facing southeast.	Minimum stand separation required for simultaneous push-back and startup for Code C aircraft on Apron 1 is 03 stands. **Exception:** If required, ATC may instruct to pushback/pull forward from stands 86 and 87 to abeam stand 85/89 on taxiway L3 to keep clear link taxiways T1 and T3 crossing	Minimum 02 stand separation shall be required to maintain between pushback aircraft and docking aircraft on Apron 1.
71L	Code C Aircraft Pushback and startup abeam stand 71L on Taxiway L4 facing southeast.	Stand 70L, 70, 70R, 71R,71, 72, 73	Stand 70L, 70, 70R, 71R, 71, 72
71R	Code C Aircraft Pushback and startup abeam stand 71R on Taxiway L4 facing southeast.	Stand 70L, 70, 70R, 71L, 71, 72, 73, 74	Stand 70L, 70, 70R, 71L, 71, 72, 73

Note: All aircraft taxing on Apron 1 to use minimum power in order to avoid jet blast.

B. Pushback Restrictions:

i) Minimum stand separation required for simultaneous pushback and startup of aircraft on Apron 1:

a) 03 stands separation shall be required up to code C aircraft.

b) 03 stand separation shall be required for code D and E aircraft if the aircraft in front of it is up to code C.

Example

Pushback from	Pushback permitted on
Stand 72	Stand 75

ii) Minimum 02 stand separation shall be required to maintain between pushback aircraft and docking aircraft on Apron 1.

Example

Pushback from	Docking permitted on
Stand 72	Stand 74

Note: “U” turn from L3 to L4 and vice-versa allowed only through link taxiway’s.

C. Pushback and taxi procedure for stands 70L, 70, 70R, 71L,71, 71R 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93 and 94.

i) All stands in apron 1 are power in/push back.

ii) Stand number 70 and 71 are MARS stands (70L, 70R, 71L & 71R are code C stands and 70 & 71 are suitable for aircraft code E and D operations respectively).

iii) Stand numbers from 72 to 94 are independent code C stands.

iv) The pushback operator shall ensure that the aircraft is aligned on the taxi lane and nose wheel of aircraft should be abeam the lead-in line of the stand unless otherwise advised by ATC.

v) Ensuring correct push back on Apron 1 taxiway L4/L9/L3 (L3 and L4 are up to Code C compatible and L9 compatible for Code D/E aircraft) as per instructions received from ATC shall be responsibility of Airline operator/GHA

vi) Stand no. 70 and 71 are connected to taxiway L9.

vii) Stand no. 70L, 70R, 71L, 71R, 72 to 83 are connected to taxiway L4.

viii) Stand no. 84 to 94 are connected to taxiway L3.

ix) Crossing of taxiway from L3 to L4 and vice-a-versa is provided abeam stand no 73, 74, 93, 94 via taxilane T4 and T5 and abeam stand no 79, 81, 86, and 88 via link T1 and T3.

x) All aircraft movement on taxiway L9 shall be under Follow Me guidance.

xi) No aircraft movement permitted on taxiway L3 and L4 during the movement of any type of aircraft on taxiway L9.

xii) Pushback from stand 70L, 70R, 71L, 71R and 72 to 83 shall be on taxiway L4.

xiii) Pushback from stand 84 to 94 shall be on taxiway L3.

xiv) Simultaneous push back and start up by Code A/B/C aircraft on taxiway L4 and L3 is permitted..

xv) Only one pushback will be permitted from stand 70 or stand 71 at a time.

xvi) Follow Me service may be used for change of lane other than crossings provided, if required.

xvii) Taxiway L3 & L4 are suitable for code C and taxiway L9 is suitable for code E aircraft operations.

xviii) Taxiway L3 & L4 are designed for independent parallel Code C Aircraft operations. No operation will be permitted on taxiway L9 during the operation of taxiway L3 or L4 and vice versa.

xix) Taxiway center line L9 is painted in broken yellow colour and Code C taxiway centreline L3 & L4 painted in continuous yellow colour.

xx) MAXSPAN 36M marking shall be provided at the origin of Code C taxiway (L3 & L4).

xxi) MAXSPAN 18.5M marking shall be provided at the origin of taxiway L5.

VI. AIRPORT COLLABORATIVE DECISION MAKING (A-CDM, SARDAR VALLABH BHAI PATEL INTERNATIONAL AIRPORT, AHMEDABAD)

1. BACKGROUND

1.1 Sardar Vallabh Bhai Patel International Airport, Ahmedabad - Collaborative Decision Making (A-CDM, Ahmedabad) undertaken at Sardar Vallabh Bhai Patel International Airport is a joint programme among all airport partners –

- Air Navigation Service Provider (ATC)
- Aircraft Operators (AO)
- Airport Operator / (AOCC)
- Ground Handling Agencies (GHA)
- Central Air Traffic Flow Management Unit (CATFMU)
- Air Force Movement Liaison Unit (AF-MLU)

All the partners are required to work in close collaboration to ensure the successful operation of A-CDM, Ahmedabad.

1.2 The efficiency of the Air Transport System is highly dependent on traffic predictability. A-CDM, Ahmedabad effectively enhances predictability (this reduces buffer times for resource planning and flight times), overall efficiency and punctuality by linking and sharing of accurate and timely information amongst Aircraft Operators, Airport Operator, ATC, etc.

1.3 A-CDM application is developed in-house by Airports Authority of India.

1.4 A-CDM system processes and utilizes the following information from the various sources mentioned below:

a) AFTN:

i) Flight plan data
ii) EOBT
iii) ATD
iv) ETA
v) NOTAM
vi) METAR
vii) ADC Number

b) Airport Operations Data Base (AODB) of Ahmedabad

i) Parking stand of Aircraft
ii) Aircraft registration
iii) Boarding status of departing aircraft
iv) Actual “Off Block” Times ***(AOBT)***
v) Actual “In-Block” Times ***(AIBT)***

c) A-CDM system database

i) Arrival and Departure Taxi timings from each parking stand
ii) Runway-in-Use and Closures
iii) Hourly handling capacity of runways
iv) Priority of flights e.g. VVIP and Emergencies etc.
v) Flight schedules

1.5 A-CDM is a Common Information Sharing Platform introduced to process timely and accurate information for quick and precise decision making. A-CDM takes into consideration the ETAs of arriving aircraft, Target Off Block Times ***(TOBT)*** of departures, Runway-in-use and the handling capacity of the runway and determines the arrival and departure sequence. Based on the sequence and the parking position, it calculates the Target Take-Off Times ***(TTOT)*** & Target Start up Times ***(TSAT)***of departures and Target In-Block Times ***(TIBT)*** for arrivals.

1.6 Implementation of ACDM will result in considerable benefits to stakeholders. The various benefits of A-CDM includes are as follows:

- Improvement in efficiency of operations
- Reduction of delays to departures at holding point.
- Orderly flow of traffic.
- Reduction in wastage of Aviation fuel.
- Reduction in Carbon footprint, reduction in Environmental Pollution.
- Optimization of airport and airspace capacity
- Reduction in R/T congestion
- Enhanced efficiency and reduced workload due to reduced R/T congestion.
- Improved Situational Awareness and increased confidence amongst ACDM Partners.
- Adherence to ADC (Air Defence Clearance) by stake holders.
- Better Passenger experience.
- Optimized use of ground handling resources
- Optimized use of parking stands, gates and terminals.
- Reduced apron and taxiway congestion
- Access to current MET information.

2. TWO KEY PARAMETERS OF A-CDM, AHMEDABAD: TOBT and TSAT.

2.1 TARGET OFF-BLOCK TIME (TOBT)

2.1.1 Definition: The time that an Aircraft Operator estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up/push back immediately upon reception of clearance from the **Surface Movement Control Unit SMC (**R/T Callsign: Ahmedabad Ground). It is the real time target of ensuring readiness of an aircraft and therefore, more accurate than the static departure time mentioned in the flight plan and hence, an ideal milestone to be used by all airport partners.

2.1.2 Purpose: Air Traffic Management (ATM) based on only FPL-data is insufficient to guarantee smooth and efficient operations on the ground and in the air. The main reason for this is the incorrect updating/no updating at all, of the Estimated Off Block Time (EOBT) of the aircraft. The result is that airport partners will have an incorrect idea about the actual status of that specific aircraft. The purpose of Target Off-Block Time (TOBT) is to provide a timely, accurate and reliable estimate of an aircraft's off block time to the A-CDM, Ahmedabad partners. Accurate TOBTs enhance operations on the ground as they provide all airport partners with a clear picture of the intentions of aircraft on the ground.

2.1.3 Communication of TOBT: Aircraft Operators shall revise their EOBTs via AFTN using CHG/ DLA message or inform ARO for revised EOBT, who in turn will originate CHG / DLA message for transmission through AFTN.

2.1.3.1 In case No delay is expected: If there is no revision of EOBT, EOBT will be treated as the TOBT. For such cases there is no need of any communication from the Aircraft Operator to the ARO.

2.1.3.2 In case delay is expected:

a) When Air Defence Clearance (ADC) is valid: If delay is within the validity period of ADC, EOBT should be revised through CHG/DLA message. Optionally TOBT can be changed in ACDM interface. But if due to some reason coordination is not possible by both the above methods then EOBT can be revised by informing ARO.

b) When Air Defence Clearance ADC is <u>NOT</u> valid:

i) If delay is outside the validity period of ADC, EOBT shall be revised through CHG/DLA message or after having coordinated with MLU and obtaining the revised ADC number, by informing ARO about revised EOBT and ADC. (Option of change of TOBT will not be available in this case).

ii) ADC Validity Period: (-) 15 minutes to (+) 45 minutes of EOBT in RPL/FPL or subsequent revision of EOBT by DLA/CHG message.

<u>Note:</u> Once Target Start-up Approval Time (TSAT) is allocated by ATC at TOBT - 30 minutes, the TOBT can be updated thrice

2.1.5 Accuracy: TOBTs must be updated to an accuracy of +/- 5 minutes.

2.2 <u>TARGET START UP APPROVAL TIME (TSAT)</u>

2.2.1 Definition: The time is calculated by A-CDM application taking into account TOBT, CTOT* and/or the traffic situation that an aircraft can expect to start-up/push back for departure.

Note: In case there is no congestion, ATC may allow the aircraft to start, where Actual Start-up Approval Time (ASAT) will be before the TSAT.

2.2.2 Purpose: TSATs provide an optimized start-up sequence, leading to Pre-Departure sequence. TSATs reduce queuing times at the runway holding points, while ensuring optimised utilisation of the runway. ATC will continue to optimize the departure order for creating the right mix of traffic.

2.2.3 Calculation of TSAT: TSAT is calculated automatically by A-CDM Application. The TSAT is calculated by taking into account TOBT, priority of aircraft, variable taxi times (VTT), and the runway capacity. Any revision of EOBT will trigger the A-CDM application to recalculate a new TSAT for that flight as well as to re-allocate earlier TSAT to next flight which meets the requisite conditions. TSAT is displayed in various ACDM interfaces in the following colour coded information.

Blue colour: TSAT - 15 minutes up to TSAT - 5 minutes; Green colour: TSAT - 5 minutes up to TSAT + 5 minutes; Red colour: After TSAT + 5 minutes.

2.2.4 Generation of TSAT: TSAT is generated by ACDM interface at TOBT - 30 minutes.

2.2.5 Communication of TSAT: TSAT will be displayed in various A-CDM interfaces.

The AO/GHA shall advise flight crew of TSAT, displayed in the A-CDM interface.

2.2.6 Priorities for generation of TSAT:

The following aircraft will be accorded priority by A-CDM application at the time of generating TSATs:

a) VVIP flights

b) Aircraft in emergency

c) Ambulance aircraft

d) Aircraft in relief missions

e) Aircraft returning to stand after push back/taxiing out for any reason.

f) The flights, in addition to the above, that require special handling with status in item 18 of Flight plan as:

i. **ATFMX:** flight approved for exemption from ATFM measures by the appropriate ATS authority;

ii. **FFR:** fire-fighting;

iii. **FLTCK:** flight check for calibration of navaids;

iv. **HOSP:** medical flight declared by medical authorities;

v. **HUM:** flight operating on a humanitarian mission;

vi. MEDEVAC: life critical medical emergency evacuation;

vii. SAR: flight engaged in a search and rescue mission

2.2.7 Airlines/GHAs are required to monitor the A-CDM interface regularly to get information on the revised TSAT if any, in respect of their aircraft. It may please be noted that the earlier allocated TSAT may get changed due to the following reasons:

i. Change of runway

ii. Change in taxi times

iii. Revised TOBT of the subject aircraft

iv. Cancellation/ revision in TOBT of other aircraft in the departure sequence.

v. Priority handling

vi. Unusual Occurrences, etc.

vii. Change of aircraft type*

viii. Application of ATFM regulation or ATC restriction resulting in new CTOT or start up delay*

ix. Inclement Weather*

x. Bird activity*

xi. Change of route (SID)*

2.2.8 Accuracy: The TSAT will remain valid for +/- 5 minutes.

3 CO-ORDINATION WITH THE CENTRAL AIR TRAFFIC FLOW MANAGEMENT UNIT (CATFMU)*

3.1 A permanent and fully automatic data exchange with the ATFMU will be established. The data transfer will enable highly accurate early predictions of landing and departure times.

*

3.2 *Furthermore, it allows for more accurate and efficient CTOT calculations due to the use of more accurate local target take-off times.* *

4. TOBT AND TSAT BASED A-CDM, AHMEDABAD START-UP/PUSH-BACK PROCEDURES

4.1 Pre-Departure Sequence: Pre-Departure sequence is the order, in which the ATC plans the aircraft to depart from their

Gate/Stands. It should not be confused with the Pre-Takeoff sequence where ATC decides on the order in which the aircraft at holding points of the runway will depart.

4.2 Start-Up and Push-Back Procedure:

i. **The pilot will contact SMC Unit (GROUND) to request en-route clearance and SID between TSAT - 15 minutes to TSAT - 5 minutes. Blue Zone.**

ii. **The aircraft must be ready to Start-up/Push-back at TOBT and request SMC (GROUND) for Start-up/ Push-back at TSAT – 5 minutes and TSAT + 5 minutes. Green Zone.**

(Note: The system may prepone TSAT up to TOBT)

iii. In case for Schedule Flight, **"Boarding"** has not started at or before **TSAT – 5 Minutes**, allotted TSAT will be cancelled and associated slot will be allocated to next eligible aircraft.

a. A new TOBT is required from Airlines Operator.

b. On receipt of new TOBT, the flight will be re-sequenced according to new TOBT and a new TSAT with a subsequent delay will be issued.

iv. **If at TSAT +5 minutes: Red Zone, ATC has not received Start-up / Pushback request, the aircraft will lose its position in sequence.**

a. A new TOBT is required from Airlines Operator.

b. On receipt of new TOBT, the flight will be re-sequenced according to new TOBT and a new TSAT with a subsequent delay will be issued

v. ATC should normally be able to issue start-up/push-back at TSAT. Pilots will be informed by ATC of any revised TSAT if there is a delay to TSAT in excess of 5 minutes.

vi. **Taxi clearance must be requested within 5 minutes of Start-up/Pushback approval time.**

(Note: If this has not occurred, SMC (GROUND) must be notified of the extent of delay. In such cases, aircraft may lose its departure slot and a new TOBT may be required in A-CDM, Ahmedabad interface. AO shall be responsible to obtain newADC number from IAF-MLU)

vii. The **Pre-Departure (Off- block) Sequence** will be determined **in accordance with Target Start up Approval Time (TSAT) and NOT in accordance with the Start-up Request.** Pre-Departure Sequence will not have any bearing on Actual Departure sequence.

(Note: Actual Departure sequence may differ from pre-departure sequence in order to optimize the Runway and Airspace utilization.)

** [Future provision kept for incorporating with India – Central Air Traffic Flow Management]*

5 TERMS AND ABBREVIATIONS

All Acronyms are time parameters which have a standard length of four characters and have been arranged as per aircraft movement sequence.

[While developing the A-CDM, Ahmedabad procedures, the guidance material published by Euro control, ICAO A-CDM and other Organizations has been used].

S. NO.	ACRONYMS	DEFINITION	EXPLANATION
5.1	ELDT	Estimated Landing Time	The estimated time that an aircraft will touchdown on the runway. (Equivalent to ATC ETA = Estimated Time of Arrival = landing).
5.2	ALDT	Actual Landing Time	The time that an aircraft lands on a runway. (Equivalent to ATC ATA = Actual Time of Arrival = Landing, ACARS = ON).
5.3	EXIT	Estimated Taxi- In Time	The estimated taxi time between landing and in-Block.
5.4	AXIT	Actual Taxi-In Time	AXIT = AIBT – ALDT
5.5	SIBT	Scheduled InBlock Time	The time that an aircraft is scheduled to arrive at its first parking position.
5.6	EIBT	Estimated InBlock Time	The estimated time that an aircraft will arrive in-blocks. (Equivalent to Aircraft Operator's ETA = Estimated Time of Arrival).
5.7	AIBT	Actual In-Block Time	The time that an aircraft arrives in-blocks. (Equivalent to Aircraft Operator's ATA = Actual Time of Arrival, ACARS = IN).
5.8	STTT	Scheduled Turnaround Time	STTT = SOBT – SIBT
5.9	ETTT	Estimated Turnaround Time	The time estimated by the AO/GH on the day of operation to turnaround a flight taking into account the operational constraints.
5.10	MTTT	Minimum Turnaround Time	The minimum turnaround time agreed with an AO/GH for a specified flight or aircraft type.
5.11	ATTT	Actual Turnaround Time	ATTT = AOBT – AIBT
5.12	ASRT	Actual Start Up Request Time	Time the pilot requests start up clearance.
5.13	TSAT	Target Start Up Approval Time	The time provided by ATC taking into account TOBT, CTOT* and/or the traffic situation that an Aircraft can expect start-up / push back approval. Note: The actual start up approval (ASAT) can be given in advance of TSAT.
5.14	ASAT	Actual Start Up Approval Time	Time that an aircraft receives its start-up approval.
5.15	SOBT	Scheduled OffBlock Time	The time that an aircraft is scheduled to depart from its parking position.
5.16	EOBT	Estimated OffBlock Time	The estimated time at which the aircraft will start movement associated with departure (ICAO)
5.17	TOBT	Target OffBlock Time	The time that an Aircraft Operator estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up / push back immediately upon reception of clearance from the SMC (GROUND).
5.18	AOBT	Actual OffBlock Time	Time the aircraft pushes back / vacates the parking position. (Equivalent to Aircraft Operator's ATD = Actual Time of Departure & ACARS = OUT)
5.19	ARDT	Actual Ready Time (for Movement)	When the aircraft is ready for start-up/push back or taxi immediately after clearance delivery, meeting the requirements set by the TOBT definition.

5.20	EXOT	Estimated TaxiOut Time	The estimated taxi time between off-block and take off. This estimate includes any delay buffer time at the holding point prior to take off.
5.21	AXOT	Actual Taxi-Out Time	AXOT = ATOT – AOBT
5.22	ETOT	Estimated Take Off Time	The estimated takeoff time taking into account the EOBT plus EXOT.
5.23	CTOT*	Calculated Take Off Time*	A time calculated and issued by the appropriate Central Management unit, as a result of tactical slot allocation, at which a flight is expected to become airborne. (ICAO Doc 7030/4 – EUR, Table 7)*
5.24	TTOT	Target Take Off Time	The Target Take Off Time taking into account the TOBT/TSAT plus the EXOT. Each TTOT on one runway is separated from other TTOT or TLDT to represent vortex and/or SID separation between Aircraft.
5.25	ATOT	Actual Take Off Time	The time that an aircraft takes off from the runway. (Equivalent to ATC ATD = Actual Time of Departure, ACARS = OFF).

VII. All ACFT TAX APN 1, 2 & 3 to use minimum power in order to avoid jet blast.

VIII. Note (Aircraft Stands 31 to 43):

i. Stand 35 is independent up to code C stand.

ii. Stop-line marking for these stands is marked as per aircraft type.

iii. Stand 31A, 31L, 31, 31R, 32L, 32, 32R,33, 34, 34R, 35,36L, 36, 36R, 37, 38, 39, 40, 41, 42, 43 are provided with AVDGS and available for operations H24.

iv. All stands have stand – identification marking painted on the stand as well as on the stand lead- in line.

v. All stands are provided with stand identification sign and coordinates.

vi. To enhance usability, MARS (Multiple Aircraft Ramp system) stands have been provided with multiple marking of stand lead in line, wherein main parking stand is painted in continuous yellow colour and 'L' and or 'R' stand painted in broken yellow colour.

IX. Note (Aircraft Stands 70 to 94):

i. Stand 70 and 71 are multiple aircraft ramp system (MARS).

ii. Stands 70 is suitable for parking aircraft up to code E.

iii. Stand 71 is suitable for parking of aircraft up to code D.

iv. Stand 70 to 94 (except 70 and 71) are suitable for parking of aircraft up to code C.

v. Stop-line marking for these stands are marked as per aircraft type.

vi. To enhance usability, MARS (Multiple Aircraft Ramp System) stands have been provided with Multiple lead in line, wherein main stands are painted continuous yellow, and 'L' and 'R' stands are painted in broken yellow.

vii. Stands 77, 78, 79, 80 are provided with AVDGS. These AVDGS are available for Operations H24.

viii. All stands are provided with stand identification signs and with WGS coordinates.

X. PARKING STAND RESTRICTION: Parking restrictions in respect of the Apron 2 stands are as tabulated below:

Parking on Stand	Restrictions of parking on adjacent stand	Parking on Stand	Restrictions of parking on adjacent stand
31A	No restriction	35	No restriction
31L	31	36L	
31	31L, 31R	36	
31R	31	36R	
32L	32	37	No restriction
32	32L, 32R	38	No restriction
32R	32, 33	39	No restriction
33	32R	40	No restriction
34L	34	41	No restriction
34	34L, 34R	42	No restriction
34R	34	43	No restriction

VAAH AD 2.21 NOISE ABATEMENT PROCEDURES

Consistent with safety of aircraft operations and in consideration of high intensity runway operations, pilots should minimise the use of reverse thrust after landing to reduce disturbance in areas adjacent to the aerodrome.

VAAH AD 2.22 FLIGHT PROCEDURES

I. Departure aircraft passing 1000 FT change over to Ahmedabad Approcach (Frequency 119.8 MHz) unless otherwise specified by ATC Tower.

Arrivals after vacating the RWY CTC Ahemdabad GND 121.675 MHz. If unable contact 118.100 MHz.

II. ADS-B based ATS Surveillance services are provided to suitably equipped aircraft on opportunity basis within the coverage area of Ahmedabad ADS-B sensors in the jurisdiction of Ahmedabad Area Control and Approach Control. ADS-B equipped aircraft flying within the jurisdiction of Ahmedabad Area Control and Ahmedabad Approach Control to comply with relevant provisions.

III. RNAV(GNSS/DME-DME-IRU) SIDS AND STARS

1. INTRODUCTION

1.1 For RNAV(GNSS/DME-DME-IRU) SIDs and STARs operations, the aircraft shall be GNSS/DME-DME-IRU equipped and the navigation systems shall meet ICAO RNAV-1 standard of accuracy or equivalent.

1.2 Operators/pilots who are not approved to fly the RNAV(GNSS/DMEDME-IRU) SIDs and STARs shall fly the conventional SIDs or expect radar vectors from ATC.

2. ARRIVALS

2.1 STARs are presented in diagrammatic format on a chart which comprises two main elements:

a) A TRANSITION route; and

b) An ARRIVAL route.

2.2 A TRANSITION starts at a waypoint on the ATS route and then requires VOR/DME or RNAV tracking to position the aircraft for the ARRIVAL route.

2.3 All arriving aircraft are required to follow the appropriate TRANSITION and ARRIVAL routes.

2.4 Additional elements on the STAR chart include the following:

a) Vertical restrictions - designed to contain aircraft in controlled airspace and to separate aircraft from obstacles and to avoid, to the degree possible, conflict with departing traffic.

b) Speed restrictions - designed for flow control purposes.

c) Minimum safe altitude (MSA) - provides a minimum clearance of 1000Ft above all objects located in an area contained within 25NM of AAE VOR.

2.5 Arrivals can expect radar vectors to intercept the localizer for an ILS approach from the RNAV STARs.

2.6 STARs shall be issued by ATC in the following order:

a) ARRIVAL identifier;

b) TRANSITION identifier;

c) Runway-in-use;

d) An assigned level

2.7 Holding

a) AH393 – 1 min left hand race track pattern, inbound track 215 deg M, minimum holding altitude 9000Ft

b) AH394 – 1 min right hand racetrack pattern, inbound track 212 deg M, minimum holding altitude 9000Ft

c) AH367 – 1 min right hand racetrack pattern, inbound track 315 deg M, minimum holding altitude 9000Ft

d) AH374 – 1 min left hand race track pattern, inbound track 345 deg M, minimum holding altitude 4000Ft

3. DEPARTURES

3.1 All departing aircraft will be cleared on the appropriate RNAV(GNSS/DME-DME-IRU) SID.

3.2 A Transition will be issued by ATC in conjunction with the RNAV(GNSS/DME-DME-IRU) SID

4. TRANSITION

4.1 Aircraft may be radar vectored off a Transition/RNAV SID/RNAV STAR. Such aircraft will subsequently be given an instruction to intercept the appropriate Transition/RNAV SID/RNAV STAR.

5. VERTICAL AND SPEED RESTRICTIONS

5.1 Pilots shall comply with an ATC assigned level. Pilots shall also adhere to the vertical and speed restrictions depicted on the cleared Transition and RNAV(GNSS/DME-DME-IRU) SIDs/STARs. ATC clearance will take precedence when the ATC clearance does not allow the pilots to adhere to the vertical and speed restrictions depicted on the Transition and RNAV(GNSS/DME-DME-IRU) SIDs/STARs.

6. OPERATORS' PROCEDURES

6.1 Pilots shall inform ATC when on-board equipment does not meet the requirements of RNAV. Pilots can then expect radar vector from ATC.

RADIO COMMUNICATION FAILURE PROCEDURES
SARDAR VALLABHBHAI PATEL INTERNATIONAL AIRPORT
AHMEDABAD (VAAH)

1.INTRODUCTION

Aircraft emergencies encompass a wide range of critical situations that can threaten the safety of the flight, its occupants, or people on the ground. Among various emergencies, Radio Communication Failure (RCF) is one of the common occurrences which may arise during any phase of flight. Understanding and responding to RCF situations effectively is crucial for pilots and air traffic controllers to mitigate risks and ensure the best possible outcome.

The Radio Communication Failure may occur due to:

•Airborne Transmitter and/or Receiver Failure

•Ground Station Transmitter and/or Receiver Failure

The objective of this document is to standardize the actions to be taken by the pilot of arriving and departing aircraft experiencing radio communication failure at Ahmedabad Airport.

GENERAL:

i)All Transponder equipped aircraft experiencing Radio Communication failure shall set transponder to Mode A/C code 7600 as soon as practicable.

Note: This Requirement of Setting transponder to Mode A/C code 7600 in no way imposes any restriction on the pilot's decision to set transponder to Mode A/C code 7500 or 7700, whenever required.

ii)Watch for ALDIS Lamp signal shown by ATC, when established on final approach of assigned runway.

iii)In case communication failure takes place prior to establishing on final approach track, maintain the last assigned level or FL070 whichever is higher and proceed direct to AAE VOR (113.100 MHz) to join AAE holding pattern. Descend to 2000 feet in hold. After reaching 2000 feet, carry out instrument approach procedure via overhead AAE.

iv)In case radio communication failure takes place after establishing the final approach track, aircraft may continue the approach and land if runway visual or initiate go around climbing to 2000 feet then turn right to join AAE VOR (113.100) hold and carry out instrument approach procedure via overhead AAE.

v)If AAE VOR is not available, aircraft shall hold over Ahmedabad ARP (230416N 0723735E) in 1-minute right hand pattern, descend in the hold to 2000 feet. Reaching 2000 feet leave the hold to establish final approach for the assigned runway or carry out visual approach for the assigned runway.

vi)Aircrafts shall keep looking for potential traffic while executing any manoeuvre as specified in this document.

vii)An aircraft equipped with ADS-B may indicate the loss of air ground communication by all of the available means.

viii)The pilot of affected aircraft may try to communicate to the concerned ATS unit through 121.5 MHz or the frequency of adjacent units/stations, or by relay through another aircraft.

ix)Pilot may make blind transmissions to ATC of all necessary reports and actions taken by the aircraft, e.g., descent, turn, proceeding to waypoint etc., irrespective of whether partial/ complete RCF has been experienced or not.

x)Aircraft may be suffering from partial radio communication failure, e.g., aircraft may be able to receive instructions from ATC and following the same.

xi)To ascertain RCF in an aircraft, ATC should:

a)Change aircraft heading by 30 degrees or more

b)Instruct to change SSR code/squawk ident, and

c)Observe the compliance by pilot of above instructions.

xii)If it is identified by ATC using the procedure above as mentioned in para (xi), that the aircraft is able to receive their transmissions, then ATC shall continue providing suitable ATC instructions.

xiii)After landing, aircraft shall lookout for follow-me jeep on taxiway Alpha holding point (Runway in use-23) and taxiway R4 holding point (Runway in use-05) and after sighting the Follow-me Jeep, Aircraft shall follow the Follow Me Jeep for parking to the allotted bay. Upon sighting the Marshaller, aircraft shall stop following "Follow me Jeep" and shall follow marshaller for parking

3. ASSIGNED RWY AND ITS AVAILABILITY FOR RCF AIRCRAFT

i)If arriving aircraft has been advised and acknowledged of runway in use at Ahmedabad, then such runway shall be considered as the assigned runway.

ii)If arriving aircraft has not been advised of runway in use at Ahmedabad, then Runway 23 shall be considered as assigned runway.

iii)In strong tail wind conditions for assigned Runway, aircraft experiencing RCF will carry out a missed approach on assigned runway and after following complete missed approach for such approach, aircraft will carry out published instrument approach procedure for the opposite runway.

iv)In case of departures from Ahmedabad In case of departures from Ahmedabad returning due to any reason including diversion becomes RCF, the runway in use at the time of departure shall be considered as assigned runway for landing.
v)Runway lights and Approach lights in 'SWITCHED ON' position shall indicate the availability of such Runway for aircraft experiencing RCF. Irrespective of visibility/weather conditions, Runway and Approach lights in 'SWITCHED OFF' position shall indicate non- availability of such Runway for aircraft experiencing RCF.
4.PROCEDURE
i)Arriving aircraft - STAR assigned
a)Squawk 7600 or suitable ADS-B emergency/urgency message.
b)Except when descent clearance has already been received from ATC, pilot shall not commence descent before 100 NM from 'AAE VOR'. Aircraft shall continue on assigned STAR maintaining last assigned
level for 3 minutes. This allows ATC to resolve any possible traffic conflict. After 3 minutes, the affected aircraft shall proceed directly to AAE VOR maintaining/descending to cleared flight level or FL070 whichever is higher and hold overhead AAE VOR as published.
c)Descend to 2000 feet in AAE VOR hold, reaching 2000 feet Leave 'AAE VOR' to carry out published instrument approach procedure via overhead AAE VOR for assigned runway.
ii)Arriving aircraft - STAR not assigned
a)Squawk 7600 or suitable ADS-B emergency/urgency message.
b)Continue on ATS route, or last assigned heading, maintaining/descending to cleared flight level or FL070 whichever is higher. At 30 miles to 'AAE VOR', take a turn to proceed direct to 'AAE VOR' descending to/maintaining FL070.
c)If RCF takes place within 30 NM of 'AAE VOR', continue on ATS route and at 10 miles to 'AAE VOR' take a turn to proceed direct to 'AAE VOR' maintaining last assigned level or 2000 feet whichever is higher.
d)Descend to 2000 feet in AAE VOR hold (if higher) and reaching 2000 feet, leave 'AAE VOR' to carry out published instrument approach procedure via overhead AAE VOR for assigned runway.
iii)Arriving `being RADAR Vectored
a)Aircraft being radar vectored for approach, on experiencing RCF, shall maintain last assigned level and heading for 3 minutes after detecting RCF or selecting MODE A/C 7600, whichever is later, and then take a convenient turn to proceed direct to 'AAE VOR' descending/maintaining last assigned level, or FL 70 whichever is higher.
b)After crossing 30 NM inbound from 'AAE VOR'. Descend to 2000 feet, join AAE VOR hold as published.
c)If arrival becomes RCF after crossing 30NM inbound from "AAE VOR", Descend to 2000 feet, join AAE VOR hold as published.
d)Reaching 2000 feet, leave 'AAE VOR' to carry out published instrument approach procedure via overhead AAE VOR for assigned runway
iv)Departing aircraft intending to continue to its destination
a)Squawk 7600 or suitable ADS-B emergency/urgency message and continue on assigned SID or heading climbing to or maintaining cleared level or FL70, whichever is higher for 3 minutes.
b)After 3 minutes of setting Mode A/C code 7600 or reaching FL 70 or cleared Flight level (if higher than FL70) whichever is later,
I)continue on SID to join ATS route and climb to filed Flight level and continue as per the filed flight plan, or
II)if following Radar heading, turn to join ATS route by shortest route maintaining cleared flight level/FL 70.
After joining ATS route, climb to filed Flight level and continue as per the filed flight plan.
v)Departing aircraft intending to land back at Ahmedabad
a)Squawk 7600 or suitable ADS-B emergency/urgency message and continue on assigned SID or heading climbing to or maintaining cleared level or FL70, whichever is higher for 3 minutes.
b)After 3 minutes of setting Mode A/C code 7600 or reaching FL 70 or cleared Flight level (if higher than FL70) whichever is later,
I)take a turn to proceed directly to AAE VOR descending to/maintaining FL070, or
II)If fuel jettisoning is required, take a turn to proceed directly to location bounded by R305-R330 between 35NM-55NM AAE VOR, climbing to/maintaining FL070 for fuel jettisoning.
III)Upon completion of fuel jettisoning, aircraft shall take a turn to proceed directly to AAE VOR maintaining FL070.
c)After crossing 30 NM inbound from 'AAE VOR'. Descend to 2000 feet, join AAE VOR hold as published.
d)Reaching 2000 feet, leave 'AAE VOR' to carry out published instrument approach procedure via overhead AAE VOR for assigned runway
vi)For VFR flights
a)Squawk 7600 or suitable ADS-B emergency/urgency message.
b)VFR flights on experiencing RCF, shall maintain last assigned altitude and fly on course to Ahmedabad and then take a convenient turn to join right/left downwind of the assigned runway. VFR flights shall keep looking for any traffic and land on the assigned runway

VAAH AD 2.23 ADDITIONAL INFORMATION

1. Taxiways:						
S.No	**TWY Designation**	**Location**	**PCR**	**Length (M)**	**Width (M)**	**Shoulder (M)**
1	TWY A	Beginning of RWY05, towards North	943/F/B/W/T	308.25	23M, Flexible	10.5
2	TWY B	186 m from Beginning of RWY 05, towards North	862/F/A/W/T	337	23M, Flexible	10.5
3	TWY C	2545 m from RWY 23 Threshold, Northwest of RWY	1181/F/C/W/T	297	23M, Rigid	10.5
4	TWY D	1711 m from Beginning of RWY 23, towards North	862/F/A/W/T	320	23M, Flexible	10.5
5	TWY F	Link opposite TWY B south of RWY 05/23 connecting. Isolated aircraft parking position at 188m from beginning of RWY05	1354/R/D/W/T	262	23M, Rigid	10.5
6	TWY G	Link connecting Apron 2 with TWY P	1181/F/C/W/T	152.25	23M, Rigid	10.5
7	TWY H	Link connecting Apron 2 with TWY P	1181/F/C/W/T	152.25	23M, Rigid	10.5
8	TWY K	Beginning of RWY 05 on the south side connecting Airforce dispersal	185/R/D/W/T	--	18.5M, Rigid	3.1
9	TWY L1	--	1506/R/B/W/T	175	23M Concrete	Suitable for up to Code E operations and Connecting from TWY A, B to Apron 1 facilities and vice versa.
10	TWY L3	--	1506/R/B/W/T	770	23M Concrete	Suitable for up to Code C operations and connecting southwestern stands of Apron 1 to TWY L1, TWY A and TWY B.
11	TWY L4	--	1506/R/B/W/T	770	23M Concrete	Suitable for up to Code C operations and Connecting Northeastern stands of Apron 1 to TWY L1, TWY A and TWY B.
12	TWY L5	--	1506/R/B/W/T	85	10.5M, Concrete	Connecting TWY L4 to Apron 3.

13	Taxilane L6	--	1175/R/A/W/T	133	--	Connecting to Apron 3 stands 25 to 30 from TWY L5
14	Taxilane L7	--	1175/R/A/W/T	133	--	Connecting all apron 3 stands for taxi-out only
15	Taxilane L8	--	1175/R/A/W/T	133	--	Taxilane L8 is connected to TWY L5 with all aircraft stands between 25A to 30 A.
16	TWY L9	--	1506/R/B/W/T	770	23M Concrete	Suitable for code D/E aircraft operation and connecting wide body stands to TWY L1, TWY A and TWY B.
17	TWY M	Link connecting via TWY K and Airforce dispersal	185/R/D/W/T	--	19.1M, Rigid	N/A
18	TWY P	TWY P located Parallel to RWY BTN TWY A to R4	1181/F/C/W/T	1850	23M Flexible	10.5
19	TWY R4	1621 m from RWY 23 Threshold, Northwest of RWY 05/23	1181/F/C/W/T	157	23M Flexible	10.5
20	Link T1	--	1506/R/B/W/T	--	--	Linking TWY L3 and L4
21	Link T3	--	1506/R/B/W/T	--	--	Linking TWY L3 and L4
22	Link T4	--	1506/R/B/W/T	--	--	Linking TWY L3 and L4
23	Link T5	--	1506/R/B/W/T	--	--	Linking TWY L3 and L4
24	TWY R1	3011 M from RWY 05 Threshold, Northwest of RWY	776/F/C/W/T	158	18 M Flexible	3.5 M
25	TWY R	TWY R located Parallel to RWY BTN TWY R4 to R1A connecting to TWY P	776/F/C/W/T	1655	18 M Flexible	3.5M
26	TWY R3	1960 m from Threshold of RWY 05	776/F/C/W/T	300	18M, Flexible	3.5M
27	TWY R5	1960 m from Threshold of RWY 23.	900/F/C/W/T	351	45M, Flexible	7.5M

28	TWY R1A	Beginning of Runway 23 towards North.	776/F/C/W/T	158	18	3.5

2. Routine Wx. Radar observations taken at 0150, 0250, 0550, 0850, 1050, 1350, 1450, 1750, 2050 and 2250. during pre monsoon, monsoon and post monsoon seasons in case of no echoes. In case echoes are observed additional observations are taken at 0350, 0450, 0650, 0750, 0950, 1550, 1650, 1850, 1950 and 2150. Observations are also taken in other seasons during bad Wx. Conditions.

3. Obstacle Light provided on lightening arrestor pole on Antenna of the ASR/MSSR building antenna height 237 FT Coordinates 230412N 0723754E distance from centerline of RWY - 870 FT south of RWY, distance from beginning of RWY 05-3350 FT.

4. .All arriving aircraft (Code C) from RWY 23 expect to vacate via TWY C LDA 2545 Meters.

5. Turn pad for intermediate DEP RWY23 is available for Code"C" Acft A321/B737. The details of this turn pad is as below:

i. Location: 762M from beginning RWY23 or 900M NE from TWY D intersection with RWY.

ii. Surface: Bituminous

iii. PCR: 862/F/A/W/T

iv. Light: Edge Light

v. Marking: Yellow Colour Turn Pad Marking

vi. Critical aircraft: Code ‘C’ A321/B737

vii. For departure from intermediate turn pad RWY23 following distances are available -

TORA (M)	TODA (M)	ASDA (M)
2743	2743	2743

viii. Departure from this intermediate turn pad can be initiated by ATC as intermediate departure RWY 23 with concurrence of pilot or requested by pilot.

6. Intersection Take Off Run Available as follows:

RWY DESIGNATION	TWY intersection	TORA
RWY 23	TWY D	1794 M
RWY 23	TWY C	960 M
RWY 05	TWY B	3319 M
RWY 05	TWY C	2545 M
RWY 05	TWY D	1711 M
RWY 23	TWY R4	1884 M
RWY 05	TWY R4	1621M
RWY 23	TWY R1	3011 M

7. Btn 0130-1730 daily(GND to 550FT/167M AMSL), tethered helium balloon operation near Ahmedabad
i. Location: 23 00 25.56806N 072 35 51.98878E. 205 radial 4NM from AAE VOR.
ii. Colour and marking: white with blue orange and black markings.
iii. Lighting: fitted with anti collision light.
iv. Size: diameter of the balloon 22.28M
v. Maximum height of the balloon 110.253M AGL.
vi. Maximum top elevation 166.612M AMSL.
vii. Maximum drift of the balloon 30M either side.
viii. Pilots to exercise caution.

8. ADS-B Reception Frequency 1090 MHz available.

9. LOW VISIBILITY PROCEDURES :

9.1 BACKGROUND:

Until the latest amendment of DGCA Civil Aviation Requirements (CAR) Section 8, Series 'C', Part-1 on All-weather Operations, low visibility procedures were required at aerodromes for the purpose of ensuring safe operations during categories 2 and 3 approaches and/or low visibility take-offs (LVTO). However, in latest amendment to CAR (Rev. 10) Para 5.3 following provision regarding low visibility procedures is added. 'An operator shall not conduct take-off with RVR/ visibility less than standard category 1 conditions of 550M RVR/800M visibility unless low visibility procedures are enforced.' This provision necessitated the need of low visibility procedures for accommodating/permitting departures in visibility/RVR less than 800M/550M even at such airports where there are no CAT-II and CAT-III operations. Accordingly, low visibility procedures have been developed for Sardar Vallabh Bhai Patel International Airport, Ahmedabad to accommodate/permit departures in visibility/RVR less than 800M/550M from RWY 23 (runway served with RVR instruments).

9.2. DEFINITIONS:

9.2.1 Low Visibility Procedures (LVP): Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

Note: as per para 5.3 of CAR on All Weather Operations, an operator shall not conduct Take-off with RVR/Visibility less than standard CAT-I conditions of 550m RVR/800m Visibility unless low visibility procedures are enforced.

9.2.2 Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

9.2.3 Runway Visual Range: The range over which the pilot of an aircraft on the centerline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline.

9.2.4 Aerodrome Operating Minima: The limits of usability of an aerodrome for:

a. take off, expressed in terms of runway visual range and / or visibility and, if necessary, cloud conditions.

b. landing in 2D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and

c. landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height(DA/H) appropriate to the type and/or category of the operation.

9.2.5 Visibility: Visibility for aeronautical purposes is the greater of: a. The greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;

b. The greatest distance at which lights in the vicinity of 1,000 candelas can be seen and identified against an unlit background.

Note 1. — The two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).

Note 2. — The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

9.2.6 Abbreviations:

ADC : Aerodrome Control

ARFF : Airport Rescue and Fire Fighting Services

ATC : Air Traffic Control

ATM : Air Traffic Management

CFT : Crash Fire Tender

ATIS : Automatic Terminal Information Service

DG : Diesel Generating Set

LVP : Low Visibility Procedures

MET : Meteorology

RWY : Runway

RVR : Runway Visual Range

SMC : Surface Movement Control

SP : Safeguarding Procedures

TDZ : Touchdown Zone

TWR SUP : Tower Supervisor

TWY : Taxiway

WSO : Watch Supervisory Officer

9.3. GENERAL:

The Low Visibility Procedure (LVP) incorporates safeguarding measures to mitigate runway incursions and defines operational restrictions to ensure safe Airside Operations taking into account the available Aerodrome facilities.

9.4. MINIMUM REQUIREMENTS:

The following Aeronautical Ground lights and RVR equipment shall be serviceable to the required standard to support Low Visibility Procedures:

a. Runway edge lights,

b. Runway end lights,

c. Real time TDZ RVR.

d. Stand by Power supply to maintain switch over time of 1 Second for Runway

Edge Lights and Runway End Lights. This requirement can be met with the help of DG Set and/or UPS.

9.4.1 Unserviceability of Aeronautical Ground Lights/ Equipment before Implementation of LVP.

Low Visibility Procedures will not be implemented when any of the light/equipment mentioned in para 4 above is un-serviceable or is not maintained as per the required standard.

Aeronautical Ground Lighting Facility	Un-serviceability	Restrictions

Runway Edge lights	More than 15% of all lights are unserviceable	LVP operations will be suspended.
	Any two consecutive lights or more are unserviceable	
Runway End lights	More than 15% of all lights are unserviceable	LVP operations will be suspended.
	Any two consecutive lights or more are unserviceable	
Standby Generators/UPS	Any of the generator/UPS is unserviceable	LVP operations will be suspended.

9.4.2 Unserviceability of Aeronautical Ground Lights/ Equipment after Implementation of LVP.

When any of the light/equipment mentioned in para 4 above becomes un-serviceable or fails to meet the required standard during periods of LVP, TWR/SMC shall advise the aircraft accordingly and LVP shall be suspended and information to this effect shall be included in ATIS broadcast.

9.5. SAFEGUARDING PROCEDURES:

Safeguarding Procedures (SP) are instructions for relevant airport, departments and airside operators to prepare ground services and facilities for low visibility operations in order that when LVP are implemented all safeguarding procedures are complete. Duty Officer Tower/Tower Supervisor will initiate and co-coordinate with all the concerned agencies for completion of safeguarding procedures before implementation of Low Visibility Procedures.

9.5.1 Safeguarding Procedures shall be initiated when:

a. The Visibility/RVR is less than 1200m and visibility/RVR is forecast to deteriorate to 800m or less; and/or

b. The cloud ceiling is less than 400ft and forecast to fall to 200ft or less.

9.5.2 Safeguarding procedures include:

a. Positioning of 1 CFT each at the two predetermined positions at Glide path RWY 23 and at fire approach road near TWY Delta.

b. Stopping of all maintenance works on the manoeuvring area as well as removal of all men and mobile equipment from the said area.

c. Ensuring availability of secondary power supply for change over time of maximum one second for RWY Edge and RWY End lights supported by UPS.

(**NOTE:** RWY Edge and Rwy End lights may continue to operate on main power supply during safeguarding Procedures. Whenever, LVP is to be implemented as per para 6 below, the RWY Edge and RWY End lights shall be put on Standby Power Supply (DG set or UPS). This operation need to be completed before LVP is implemented.
(As UPS is available at Sardar Vallabh Bhai Patel International Airport, Ahmedabad and is capable of maintaining the required AGL system (refer table under para 4.1) with one second of Switch Over time with Main Supply, the main supply can continue to be primary supply and the Generator Supply can be kept as Standby Power supply. In case of UPS is unserviceable, Generator supply will become primary source of power supply and Main power supply shall act as standby power supply.)

d. The appropriate Aeronautical ground lights must have been inspected during the hour preceding implementation of LVP, and thereafter once every two-hour period. These lighting inspections should be accorded priority and, if necessary, aircraft operations may be delayed.

9.6. LOW VISIBILITY PROCEDURES:

9.6.1 Implementation of Low Visibility Procedures:

SMC shall inform Duty Officer Tower controller whenever Visibility/RVR reduces to 800 Meters or below and/ or cloud ceiling is at 200 ft or below. Tower supervisor shall coordinate with all the agencies to confirm whether the Safeguarding procedures have been completed or not. When Visibility/RVR falls below 800m/550M and or Cloud Ceiling is 200 ft or below and safeguarding procedures are complete, Tower supervisor will implement Low Visibility Procedures. Duty Officer Tower shall inform all users of the imposition of low visibility procedures.

9.6.2 Action by various units during LVP:

a. Duty MET Officer shall keep Duty Officer Tower informed of any change in Visibility/ RVR.

b. SMC shall ensure that the towing of aircraft is done under escort of “Follow Me” vehicles. “Follow Me” shall follow the route cleared by ATC;

c. SMC shall not permit any ground run on the manoeuvring area except idle power run on the stands;

d. SMC shall ensure that “Follow Me” services are provided to pilots on request;

e. The number of the vehicles on the manoeuvring area shall be restricted to bare minimum and records of all vehicles operating on the manoeuvring area shall be maintained by SMC.

f. The following may be included in ATIS. “**LOW VISIBILITY PROCEDURES IN FORCE**”.

g. TWR shall permit departures only from the beginning of the Runway in use.

h. Whenever visibility/ RVR is less than 800/550M, duty officer tower shall confirm from pilot that the reported RVR value is within minima before issuing take-off clearance.

i. In-Charge Electrical shall continuously monitor the main and Standby Power supply to ensure change over time of maximum one second for RWY Edge and RWY End lights during low visibility operations and report any unserviceability to Tower immediately

j. No departure shall be permitted below RVR of 400M due to non-availability of RCL.

9.7. TERMINATION OF LOW VISIBILITY PROCEDURES:

a. When Visibility/RVR improves to 800M/550 M or more and cloud ceiling is 200 feet or higher and trend is for improvement, Tower Supervisor/Duty Officer Tower would terminate operations of LVP. He may obtain advice from Duty Met. Officer regarding improvement in weather conditions before the termination of LVP.

b. Duty officer Tower shall inform SMC/ARFF/In-charge electrical-Engg. regarding the termination of LVP operations.

c. On cancelling of LVP, following message shall be included in two subsequent ATIS broadcasts. “**LOW VISIBILITY PROCEDURES CANCELLED**”.

d. If SP are implemented and LVP are not subsequently implemented and the visibility/RVR improves and is more than 1200 m and/or the cloud ceiling is 400ft or higher and both are forecast to remain above the required SP criteria, Tower Supervisor/ Duty Officer Tower may cancel SP.

9.8. ACTIONS BY OTHER AGENCIES (AIRLINES, REFUELLING COMPANIES, CATERING AGENCIES, ETC.)

a. Every year before commencement of monsoon/winter season, a meeting will be held by Airport Director, to inform all airlines and agencies operating at airport about their roles/responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.

b. All the agencies shall ensure that staff and drivers are suitably trained during Low Visibility operations.

c. A refresher program for ATCO’s and personnel responsible for airside operations shall be conducted every year.

d. All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations operate in the operational area during periods of low visibility. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.

e. All the vehicles must have their obstruction lights “ON” during Low Visibility Procedures operations.

f. All instructions/sign boards provided for vehicular movement area/service roads, must be followed while operating in the operational area.

10. ASMGCS commissioned to enhance visual surveillance.

11. Apron 3:-

i. Aircraft stands 25 to 30 are facing North-East and aircraft stands 25A to 30A are facing South-West.
ii. Fuelling available through bowsers.
iii. All aircraft to park under the guidance of marshaller provided by respective airlines/Ground Handling Agency.
iv. Aircraft stand identification signage and coordinates board is not available. However, surface marking (direction and location) for parking guidance has been provided for all the parking stands.
v. Simultaneous taxi out is not permitted from opposite stands.

12. Seasonal bird activity in and around airfield. In addition, bird activity observed 30 min before sunrise to two hours after sunrise and one and a half hour before sunet to 30 min after sunset. Extra caution to be observed for nocturnal birds. Pilots to exercise caution.

13. Every WED BTN 0630-0930, RWY 05/23 not available for operations due Maintenance. RWY restoration period is 30 min in case of emergency.

14. Obstacle DCWIS(Digital Current Weather Instrument System), frangible mast (10M) installed on southside of RWY23 and northside of RWY05. Obstacle is marked and lighted.

RWY	POSITION	DIST FM THR	DIST FM CL	TOTAL TOP ELEV
23	230508.95N 0723842.99E	290M	120M	66M AMSL
05	230406.39N 0723842.99E	273.77M	114.4M	64.92M AMSL

15.1.Bird scarers deployed on either sides Of RWY strip at DIST of 75m FM RCL BTN 2330-1330.
2.Men And Machinery deployed for Grass cutting 24 X 7 on RWY Strip at a distance of 75m and beyond on either sides of RCL at A distance of 45m and beyond on either sides of TWY centerline.Height of machinery restricted to 3.5m AGL.Pilot to exercise caution.

VAAH AD 2.24 CHARTS RELATED TO AN AERODROME

1	Aerodrome Chart (9 JULY 2026)
2	Aircraft Parking/Docking Chart Apron-1 (19 March 2026)
3	Aircraft Parking/Docking Chart Apron-2 (22 January 2026)
4	Aircraft Parking/Docking Chart Apron-3 (22 January 2026)
5	Aircraft Parking/Docking Chart Isolation Bay (07 August 2025)
6	Aerodrome Obstacle Cart Type- A (Operating Limitations) RWY 05(07 August 2025)
7	Aerodrome Obstacle Cart Type- A (Operating Limitations) RWY 23(07 August 2025)
8	Instrument Approach Chart -ILS or LOC Z Procedure RWY 23(9 JULY 2026)
9	Instrument Approach Chart - VOR Procedure RWY 05(9 JULY 2026)
10	Instrument Approach Chart - VOR Procedure RWY 23(9 JULY 2026)
11	Instrument Approach Chart - RNP Procedure RWY 05 (9 JULY 2026)
12	Tabular Description - RNP Procedure RWY 05 (16 May 2024)
13	Instrument Approach Chart - RNP Procedure RWY 23 (16 May 2024)
14	Tabular Description - RNP Procedure RWY 23 (16 May 2024)
15	Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 05 (13 June 2024)
16	Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 05, UUD & BODAR, 1B - Tabular Description (13 June 2024)
17	Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 05, QQZ & BVR, 1B - Tabular Description (13 June 2024)
18	Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 05, LIGPI & SASRO, 1B - Tabular Description (13 June 2024)
19	Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 23 (13 June 2024)
20	Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 23, UUD & BODAR, 1A - Tabular Description (27 NOV 2025)
21	Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 23, QQZ & BVR, 1A - Tabular Description (13 June 2024)
22	Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 23, LIGPI & SASRO, 1A - Tabular Description (13 June 2024)
23	Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 05 (13 June 2024)
24	Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 05, UUD & BODAR, 1B - Tabular Description (13 June 2024)
25	Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 05, QQZ & APANO, 1B - Tabular Description (13 June 2024)
26	Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 05, LIGPI & VASLA, 1B - Tabular Description (13 June 2024)
27	Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 23 (13 June 2024)
28	Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 23, UUD & BODAR, 1A - Tabular Description (13 June 2024)
29	Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 23, QQZ & APANO, 1A - Tabular Description (13 June 2024)
30	Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 23, LIGPI & VASLA, 1A - Tabular Description (13 June 2024)
31	RNAV1 SIDs & STARs - Waypoints 1 (13 June 2024)
32	RNAV1 SIDs & STARs - Waypoints 2 (13 June 2024)
33	RNAV1 SIDs & STARs - Waypoints 3 (13 June 2024)
34	Take Off Run Available from Intersection RWY 05(9 JULY 2026)

35 Take Off Run Available from Intersection RWY 23(9 JULY 2026)

VAAH AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

Procedure	Procedure Minima	VSS Penetration
1	2	3